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Resilience in the Arctic: Interagency Wargaming to Test Mission Continuity and Domain Awareness

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EXECUTIVE SUMMARY

A two-day, matrix-style, interagency tabletop wargame in Anchorage tested operational-level actors' ability to sustain homeland defense while responding to a natural disaster. The scenario featured a persistent volcanic ash plume from a notional Shishaldin eruption that grounded fixed-wing aviation across south-central Alaska, degraded maritime navigation, and compromised power generation through the natural gas turbine infrastructure that supplies the Anchorage region.[1] Concurrently, a Red Team representing Russia and China leveraged grey zone tactics to test the Blue Team's ability to manage homeland defense and disaster response. The gameplay generated two key findings. First, Alaskan critical infrastructure lacks redundancy in ways that are meaningfully different than most rural locations. If Anchorage is shut down, the city and the rest of the state are effectively on their own, relying on existing resources until conditions allow Anchorage to reopen. Second, a humanitarian assistance/disaster response (HA/DR) in Alaska inherently takes on a strategic nature given the state's geostrategic importance with rival nations that have the ways and means of compounding the challenges of internal crises.

Players were drawn from key operational stakeholders: the Port of Alaska, the Municipality of Anchorage Office of Emergency Management, Ted Stevens Anchorage International Airport, the Alaska National Guard, the U.S. Coast Guard, Defense Logistics Agency (DLA) Energy Americas North, Department of Homeland Security (DHS) Cybersecurity and Infrastructure Security Agency, the private sector, the Anchorage Police Department, and Alaska-based active military personnel, with white cell support from Idaho National Laboratory and the Homeland Defense Institute. The report's central claim about adaptive problem-solving capacity operating as an important pillar of state power draws on the author's separate doctoral research.[2] The wargame findings are one contribution to that broader research program.

The exercise produced a core finding that reframes how Arctic homeland defense should be assessed: Alaska has no scalable alternatives if its primary infrastructure fails, even as that architecture serves as the backbone for missile-defense support, NORAD execution, and a significant share of global air cargo. The Port of Alaska is the only Alaskan port with gantry cranes and roll-on/roll-off ramps also connected to road, rail and air hubs. Ted Stevens Anchorage International Airport is the only logistics node capable of handling the cargo volumes that sustain approximately ninety percent of the state's population. The combined capacity of every other Alaskan port could not substitute for the Port of Alaska in a sustained disruption. The state's sparse population compounds this challenge, as the infrastructure and events that surge emergency response require a level of human capital and talent that often necessitates individuals serving in multiple roles simultaneously. "Alaska is a spider web of single points of failure," one participant observed during gameplay, "and the very center of that web is the Port of Alaska."[3]

The wargame produced a second finding that one of the military participants articulated more clearly than the design itself anticipated: every HA/DR situation in Alaska now occurs within the context of strategic competition.

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[1] The eruption was modeled on the 2010 Eyjafjallajökull event

[2] Iverson (2026 anticipated) You Had One Job: Adaptative problem solving as an element of state power.

[3] CISA representative comment

The implications are significant and under-theorized. Pre-positioned supplies, pre-authorized aviation exceptions, pre-established tribal relationships, and pre-delegated legal authorities for counter-drone and cross-jurisdictional response would need to be in place before the next event begins. Anything sourced, authorized, or coordinated after the crisis starts would likely arrive too late to matter for the communities and the strategic functions most at risk.

A series of more specific insights flow from these two primary findings. Personnel absenteeism among essential responders was projected at 35-40% in a sustained event based on previous experiences of participants, driven primarily by family-care obligations that current planning frameworks treat as secondary.[4] This estimate informed the development of a game inject on day 2 requiring planners to adapt to 30% manpower shortfalls that could not be immediately reconstituted by support from the Lower 48. Critically, it was repeatedly noted that many people perform multiple roles in this small population, meaning that the loss of one person can equal the loss of capacity across multiple obligations. Additionally, a number of participants underscored that a certain number of people would leave the state entirely, leaving those roles gapped even with support in place to provide family care assistance.[5] Moreover, even if attempts to self evacuate were made, January conditions make it challenging for anyone to relocate after the port and airport shutdown. Those that make it out of Anchorage risk further "stranding themselves" and straining already tenuous logistics support for rescue across the state. The personnel finding extends beyond conventional first responders to direct labor across the full range of functions the response architecture depends on such as: longshoremen, deicing crews, snowplow operators, fuel truck drivers, turbine operators, water and wastewater operators, communications technicians, harbor pilots, and other place-bound skilled workforce categories that are not substitutable from outside the state. Alaska's small institutional community amplifies the per-individual consequence of absenteeism in ways that a contiguous United States model does not capture. The Lower 48 surge support that current planning treats as available is increasingly constrained by concurrent demands, with the winter 2026 standby notice for deployment of the Alaska National Guard to Minnesota serving as the most specific recent instance of the directional inversion of mutual aid the current operating environment produces.[6] Evacuations were not limited to personnel as aircraft sensitive to ash damage also needed to be considered. Fifth-generation aircraft likely would evacuate to the Lower 48 under stress, potentially leaving only residual air defense capability. The Cook Inlet natural gas electricity generation infrastructure on which Anchorage depends is itself vulnerable to volcanic ash through combined-cycle turbine sensitivity, which is susceptibility the wargame designers had not anticipated and that emerged through gameplay. Adversary information operations targeting specific Alaska Native communities with Russian Orthodox cultural ties exploited a vulnerability that current planning frameworks underweight or dismiss outright.

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[4] The 35-40% absenteeism figure is consistent with state Emergency Operations Plan documentation and with planning assumptions used by FEMA Region X for Alaska scenarios. The quarter-to-third population mobility figure was articulated by participants drawing on Mount Spurr evacuation patterns and on Cook Inlet earthquake response observations. Additional studies found that up to 30–40% of critical infrastructure and emergency workers faced barriers returning to duty due to family welfare concerns after Hurricane Katrina and other events.

[5] Further studies are needed to test this qualitative assessment.

6 Hernandez, J., & Lonsdorf, K. (2026, January 18). U.S. military troops on standby for possible deployment to Minnesota. NPR. https://www.npr.org/2026/01/18/nx-s1-5681372/us-military-troops-on-standby-for-possible-deployment-to-minnesota

The wargame also raised a key finding about institutional tempo. The Alaska response architecture defaults to sequential prioritization to establish command, then assess capability, then execute mission. Participants repeatedly noted that in crisis scenarios, the municipal, state, guard, and military stakeholders would not plan responses together. However, adversary gray-zone activity operates in parallel, exploiting the gaps that sequential processing creates. One of the military participants identified this finding directly in his after-action submission: information operations, command architecture, and logistics must be stood up simultaneously on Day 1, with no sequential phase in which any of them can be safely deferred. When a situation creates real tension between Title 10 and Title 32 authorities, such as a HADR response with strategic competition elements, or requires both to be engaged concurrently with limited available personnel, fissures emerge. This indicates that a surge to Alaska has to be faster and called sooner than the Lower 48; the logistical tail is simply longer and more challenging, adding an additional dimension of pressure for state and local authorities to request help immediately.

These findings have direct policy implications for investment in critical infrastructure redundancy and further testing of whole-of-society crisis response. While additional research would be needed in order to validate more specific recommendations, the criticality of the Port of Alaska and Ted Stevens International Airport would point to the need for expanding the capacity of alternate locations (e.g., the Ports of Seward, Valdez, and Homer or the Fairbanks International Airport) and improving ground lines of communication connecting population centers to these points of debarkation. This could include expanding the Alaska Railroad network, connecting Alaska to the Lower 48 by rail, improving the Alaskan Highway that connects the Lower 48 to Alaska, and investing in alternate or redundant highways between Anchorage and Fairbanks. The report also recommends distributed energy generation that reduces dependence on a single natural-gas-fired grid; direct tribal engagement during peacetime to build the relationships on which crisis response will depend; pre-crisis legal architecture work on counter-drone authorities, Stafford Act thresholds, Posse Comitatus boundaries, and tribal sovereignty; institutional reform toward parallel rather than sequential planning. Additional analysis is needed to test how a crisis would stress personnel numbers in order to inform the possibility of using/adopting workforce protection and family-support programs that extend beyond conventional responder categories to include the direct labor workforce on which response operations depend. Further research and testing is also needed to determine whether/how formal incorporation of private sector and voluntary organization arrangements into pre-positioned response capacity could be helpful. Finally, a follow-on summer-variant wargame to test the fundamentally different vulnerabilities that warm-season Alaska presents would more fully develop insights into a region that has immense seasonality in transport infrastructure and population realities and a high variance in capabilities based on weather patterns at different times of year.

Roadmap of the Report

Sections II and III set the table: the wargame's methodology, design choices, gaps in representation, and the puzzle the wargame was built to address. The puzzle is that Alaska's strategic significance — anchoring U.S. homeland defense, missile defense, NORAD execution, and the only U.S. land-border proximity to four NATO allies — is qualitatively decoupled from its population, and current analytical frameworks cannot represent the decoupling. Sections IV and V develop this: the strategic-capability and urban-vulnerability traditions that dominate U.S. assessment, why both miss the variable that matters, and how cryospheric change and the unmanned-systems evolution reshape what operating in the Arctic now requires. Section VI presents the wargame's findings. Section VII analyzes them through the adaptive-

problem-solving framework from the author's separate doctoral research. Section VIII develops early-stage recommendations. Section IX situates the findings within the 2025 National Security Strategy's Western Hemisphere reframing and argues that Alaska is the proving ground for how the United States operates in degraded environments more broadly.[7]

For more specific and nuanced insights, see Appendix F for wargame session summaries that illustrate core challenges directly from the game.

II. WARGAME DESIGN AND METHODOLOGY

The wargame was a two-day interagency matrix tabletop exercise in April 2026. The design and execution involved trade-offs, including design choices that worked as intended, design choices that produced unanticipated findings, and design choices that did not work as intended and that future iterations should approach differently.

Design rationale.

The wargame was designed to test Alaska-based operational-level actors' ability to sustain homeland defense and mass-care operations simultaneously under conditions of severe environmental degradation and adversary gray-zone activity. The design choices that followed from this objective included the selection of a volcanic eruption as the precipitating event, the use of a matrix wargame format rather than a scripted scenario, the "fight tonight/come as you are" framing for participants, and the three-move structure with end-of-move reflection products.

The volcanic eruption was selected for three reasons. First, it produces operational effects across the multiple domains the wargame needed to test (e.g., aviation, maritime navigation, supply chain, power generation, communications) without requiring the adversary to be the originator of the crisis. This stripped out the escalatory signaling that would attend a great-power-initiated scenario and let the wargame test institutional response without the framing distortions that adversary-initiation introduces. Second, it has a real precedent in Eyjafjallajökull 2010 and in the operational history of Alaska volcanism, which lets participants engage the scenario as plausible rather than fantastical.[8] Third, it produces a sustained-duration event rather than a discrete incident, which forces the wargame to test continuity of operations across multiple days rather than acute response. The selection of Shishaldin specifically, with prevailing winds projected to carry ash through the Aleutians, the Cook Inlet, and having some moderate effects toward the Fairbanks region, gave the scenario plausibility and let the design exercise multiple operational geographies simultaneously.

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[7] For the 2025 National Security Strategy's Western Hemisphere framing, see Pincus, "Protecting the North American Arctic and Beyond," and "Arctic Receives No Mention in New White House National Security Strategy, But U.S. Polar Engagement Remains," cited previously.

[8] The 2010 Eyjafjallajökull eruption in Iceland produced a sustained ash plume that disrupted European aviation for weeks, with cascading effects on supply chains and travel. The scenario's use of Shishaldin (a stratovolcano in the Aleutian Islands) was chosen for plausibility and for the prevailing wind patterns that would carry ash through the Bering Strait, the Aleutians, Cook Inlet, and the Northern Sea Route. See Alaska Volcano Observatory, "Shishaldin," at https://avo.alaska.edu/volcanoes/volcano.php?volcname=Shishaldin.

The matrix wargame format was selected because the design objective was to uncover unanticipated issues, rather than produce solutions to known problems. A scripted scenario tests how well participants execute pre-planned responses against pre-planned injects. A matrix format allows the gameplay itself to expose dynamics that neither the designers nor the participants anticipated. The format's value depends on participants being willing to engage the scenario actively rather than passively, on the white cell being willing to adjudicate unexpected developments through dice and discussion rather than predetermined outcomes, and on the red cell being willing to operate within its own strategic logic rather than executing a designer-specified sequence of actions. The wargame produced one of its most consequential single findings of the natural-gas-turbine vulnerability through exactly this dynamic. The participants articulated the vulnerability through their own operational knowledge during gameplay; the white cell adjudicated it through dice; the gameplay then incorporated the cascading effects into subsequent moves.[9]

The "fight tonight/come as you are" framing assumed that participants would arrive without significant pre-briefing on the scenario specifics, would not have rehearsed responses, and would be tested in their ability to organize coordination architecture under stress. This framing produced the slowest-moving phase of the wargame, the first half of Move 1 on Day 1, during which participants spent substantial time negotiating who would coordinate with whom, what authorities applied, and what assumptions they were operating under, which was particularly challenged by the fact that municipal, state, private sector, guard, and military respondents do not consistently practice this type of tabletop event together, as exercises traditionally remain in different silos. Several participants identified this in the after-action discussion as a design choice that worked less well than alternatives. The lead representative of DHS CISA observed during the hot wash that the room would have benefited from more upfront ground-truthing of the state of the world at the moment Move 1 began, what utilities were functioning, what declarations had been issued, what evacuation status existed, because participants spent significant time inferring conditions that in a real two-week-into-crisis scenario would have been established. What is assumed however is clear communications architecture that might not actually be representative of reality in a crisis, as observed in the 2018 San Juan experience of Hurricane Maria, where 95% of cell towers were rendered non-operational. Nevertheless, this methodological lesson is one the report acknowledges directly.[10]

The three-move structure with end-of-move reflection products worked as intended. Participants were forced to crystallize decisions, assumptions, and assessments at each phase boundary rather than allowing ambiguity to carry forward unchecked. One of the military participants identified this directly in his after-action submission as one of the design's strongest features: "The three-move structure with a defined end-of-move reflection product was highly effective. It forced the planning staff to crystallize decisions, assumptions, and assessments at each phase boundary rather than allowing ambiguity to carry forward

[9] The natural-gas-turbine vulnerability emerged during Move 1 adjudication on Day 1, when a participant from Launch Alaska working with the energy sector knowledge in the room raised the question of turbine sensitivity to volcanic ash. The white cell adjudicated through dice and through subsequent participant outreach to industry contacts.

[10] Statement by DHS CISA representative, during Day 1 hot wash. The framing that the room would have benefited from either an upfront baselining session, or more detailed scenario specification, is incorporated into the recommendations for future iterations.

unchecked. The cascading scenario design replicated the compounding friction of a real operational environment more faithfully than single-event scenarios typically do."[11]

The wargame was designed and co-led by the author and a co-lead with substantial experience in Russian and Arctic affairs. The two co-leads share professional backgrounds that include over a decade each of Department of War service and graduate scholarly training in the regional dynamics most relevant to the wargame's scenario design. The collaboration is grounded in longstanding professional relationships and complementary regional expertise. The wargame co-lead's specific contributions to the wargame execution are acknowledged here in role terms; per the Chatham House Rule applied to the wargame and the report, the co-lead's identity is not disclosed in this document.

Participant composition. The wargame's participant composition reflected the operational stakeholders who would in fact be responsible for response to the modeled scenario. Blue team participants included representatives from the Port of Alaska (Operations and Security Support, and Port Safety Coordinator), the Municipality of Anchorage (Unified Command EOC Director and Emergency Programs Manager), Launch Alaska (Deputy Director of Transportation and Chief Deployment Officer), Ted Stevens International Airport (Airport Safety Specialist), the Alaska National Guard (Joint Staff), DLA Energy Americas North (Quality Assurance Representative), the U.S. Coast Guard (Emergency Management at Sector Western Alaska and a USCG Fellow at the Ted Stevens Center for Arctic Security Studies), DHS Cybersecurity and Infrastructure Security Agency (Protective Security Advisor), and a U.S. Army Military Fellow at the Ted Stevens Center for Arctic Security Studies.

White cell support was provided by Idaho National Laboratory through their Senior Critical Infrastructure Analyst, the wargame co-lead, and the lead author. Red Team consisted of three subject matter experts drawn from academic and federal logistics backgrounds operating under collective non-attribution per the Red Team agreement.

Three originally invited participants, representatives from the University of Alaska Fairbanks International Arctic Research Center, the Alaska Army National Guard G3 Operations Staff, and a second Idaho National Laboratory analyst, were unable to attend the executed wargame. Their absence is noted in the participant roster.

Attribution convention. This exercise employed the Chatham House Rule to wargame participation. Participant statements during gameplay, during the post-game synthesis meeting, and in after-action communications are attributed by role and organization rather than by name. The convention applies uniformly to all participants except the lead author. Two participants submitted detailed after-action documents that contributed substantively to the report's framework; their contributions are quoted directly where appropriate and attributed to "one of the military participants in his after-action submission" or "one of the military participants in her after-action submission" without further identification, consistent with the protective intent of the Chatham House Rule.[12] The Red Team's collective contributions are attributed to "the Red Team" without individual identification. The white cell's collective contributions

[11] Alaska National Guard military fellow after-action submission.

[12] Full participant roster is provided in Appendix A. Non-attribution applies to individual statements made during gameplay; references in to participant statements are by role and organization unless the participant has agreed to attribution.

are attributed to "the white cell." The wargame co-lead's specific contributions during the wargame execution and post-game synthesis are attributed to "the wargame co-lead" without further identification.

The participant composition produced two significant value-adds that were deliberately designed for: The interagency mix showed coordination seams that a military-only exercise would not have exposed. One of the military participants identified this directly in his after-action submission: "The interagency audience composition created productive tension with the military planning staff and demonstrated real coordination seams that a military-only exercise would not have exposed."[13] The composition also brought operational expertise from the institutions that would actually execute a response in a real crisis. The Port of Alaska's operational practitioners, the Municipality of Anchorage's emergency management lead, the U.S. Coast Guard's operational chief, the DLA Energy logistics representative, and the others contributed direct operational knowledge rather than theoretical assessment.

The participant composition also had significant gaps that were a function of design limitation: The single largest gap was the absence of Alaska Native and tribal representation. Tribes were not in the room as players. This was a recruitment limitation rather than a design choice. Building the relationships and trust required for direct tribal participation in a wargame of this kind requires sustained engagement that the wargame's convening timeline could not accommodate. Other gaps included the absence of FEMA representation, the absence of state-level emergency management beyond the Alaska National Guard and the white cell's stand-in for the State Emergency Operations Center, the absence of the American Red Cross and other major nongovernmental humanitarian organizations, the absence of major private utilities (Chugach Electric, ENSTAR Natural Gas, Hilcorp), and limited representation from the air and maritime cargo industries. Each of these gaps shapes the conclusions the wargame can and cannot support. The wargame's most empirically grounded findings are those that bear on the institutions that were directly represented. Findings that bear on absent institutions are weaker and are reported in this document as hypotheses warranting follow-on investigation rather than as established findings.

The scenario architecture used in this wargame had been adapted once before, in a different setting and for a different purpose. The author previously developed an instructional version for a university course on the United Nations and international security, in which the players were nation-states and international organizations rather than U.S. agencies and a single state, and the unit of analysis was the international system rather than the homeland. That adaptation was a teaching exercise, not a data-generating event, and none of the findings rest on it. It is noted here because it establishes that the underlying scenario design has been run with more than one player set and has produced coherent strategic play in both.

Red Cell in good faith.

The Red Cell's representation of Russian and Chinese gray-zone activity was conducted in good faith based on the participants' best understanding of documented adversary doctrine and observed behavior. The Red Cell could not accurately predict how the Kremlin or Zhongnanhai would in fact behave in this specific scenario; however, the tabletop approach allowed the development of U.S. institutional response to adversary behavior consistent with documented playbooks. The cost-imposition finding, that ambiguous low-cost actions force disproportionate defender responses, is robust because the institutional

[13] Alaska National Guard military fellow representative, after-action submission, on the three-move structure

response is what the wargame actually measured, regardless of adversary intent. The information-operations findings, including the differential community-response dynamic and the cleared-uncleared knowledge gap exploitation, are robust because the institutional vulnerabilities the Red Cell exposed are structural features of how the U.S. response architecture currently operates.

III. THE PUZZLE

The United States is an Arctic nation that does not think of itself as one. This is not a new observation. A March 2020 study based on national online surveys conducted in late 2019 found that Americans continue to mildly disagree with the assertion that the United States is an Arctic nation with broad and fundamental interests in the region.[14] The Arctic Research Consortium of the United States has documented similar findings across multiple years.[15] The psychological-distance literature in social science provides a framework for understanding why. Construal Level Theory holds that geographically and socially distant events are mentally represented in more abstract, lower priority terms than proximate threats. The availability heuristic compounds the effect as Arctic security issues rarely dominate national consciousness unless tied to dramatic short-term events.[16] Alaska contains approximately 740,000 people. The United States contains approximately 334 million. The disproportion between Alaska's strategic significance and its demographic weight in the national political imagination has consequences for how the region is understood, resourced, and defended.

These consequences have been visible in successive national security strategy documents. The 2017 National Security Strategy declared that "great power competition has returned" and treated Russia and China as the principal challenges to American interests, but its Arctic treatment was confined to brief references to homeland defense and freedom of navigation.[17] The 2022 National Strategy for the Arctic Region elevated the region with a standalone strategy emphasizing climate change, security, sustainable development, and international cooperation through the Arctic Council.[18] The 2025 National Security Strategy dropped the standalone Arctic chapter, reframing strategic priorities around the Western Hemisphere with an explicit emphasis on near-abroad strategic primacy that, by extension, places the North American Arctic at the top of U.S. strategic concern even when it is not named.[19] Different facets

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[14] Alaska National Guard military fellow representative, after-action submission. The framing has been used as the organizing claim of the puzzle section with the author's permission.

[15] Lawrence Hamilton, Jochen Wirsing, Jessica Brunacini, and Stephanie Pfirman, "Arctic Knowledge of the U.S. Public," Arctic Research Consortium of the United States, available at https://www.arcus.org/witness-the-arctic/2017/5/highlight/2.

[16] For the foundational statement of Construal Level Theory, see Yaacov Trope and Nira Liberman, "Construal-Level Theory of Psychological Distance," *Psychological Review* 117, no. 2 (2010): 440-463. For the availability heuristic, see Amos Tversky and Daniel Kahneman, "Availability: A Heuristic for Judging Frequency and Probability," *Cognitive Psychology* 5, no. 2 (1973): 207-232.

[17] White House, *National Security Strategy of the United States of America* (December 2017). The relevant Arctic language is brief; the phrase "great power competition" appears prominently throughout the document as the principal organizing concept.

[18] White House. (2022, October). *National strategy for the Arctic region.* https://2021-2025.state.gov/national-strategy-for-the-arctic-region-nsar/

[19] For the 2025 National Security Strategy's Western Hemisphere framing, see Pincus, "Protecting the North American Arctic and Beyond," and "Arctic Receives No Mention in New White House National Security Strategy, But U.S. Polar Engagement Remains," cited previously.

of the Arctic security dynamic are being examined without arriving at a clear operational understanding of the actual challenge. The 2017 framing identified great power competition, but then treated it as a phenomenon between militaries in defined theaters, separable from homeland defense and humanitarian assistance.[20] The 2022 framing identified climate change and sustainable development and then treated security as one priority among several rather than as the operating context within which the other priorities must be pursued.[21] The 2025 framing identifies homeland defense and Western Hemisphere primacy but the institutional architecture connecting the strategic framing to Arctic operational planning remains incomplete.[22]

The wargame was designed to account for these evolving security dynamics against this institutional backdrop. Its core premise was that the gap between the United States' articulated Arctic strategy and what is required for an effective applied strategy is best highlighted by stress-testing the operational architecture needed to execute the strategy under realistic conditions. A volcanic eruption was selected as the precipitating event for methodological reasons. A volcanic eruption strips out the framing problems of an adversary-initiated crisis and lets the operational environment force decisions without the escalatory signaling that would attend a great-power scenario. The wargame is not predicting that a volcano will erupt. It is using a plausible environmental crisis to test institutional capacity that the same operational architecture would have to execute regardless of the precipitating event.

The wargame produced one finding that should reframe how Arctic homeland defense is assessed across all three of the strategy documents. The finding was articulated by one of the military participants in his after-action submission and is best stated in his words: “This was a strategic competition event with a HADR mission operating inside it.” The reframe is not that strategic competition and humanitarian response are in tension and must be balanced. The reframe is that strategic competition is the operating environment within which humanitarian response now occurs. The implications cascade through every aspect of how Arctic crises must be planned, resourced, and executed. If strategic competition is the operating environment, then information operations, command architecture, and logistics must be stood up simultaneously on Day 1 of any crisis. Sequential planning that treats strategic competition as a separate problem set to be addressed after the immediate humanitarian situation stabilizes is structurally incompatible with the operational reality. Adversaries do not wait for the humanitarian situation to stabilize before exploiting the vacuum.

The wargame demonstrated this proposition through gameplay rather than asserting it through framing. Red Team, three subject-matter experts modeling Russian and Chinese gray-zone doctrine in good faith, deliberately chose not to deploy cyber attacks during the scenario. The reasoning was that environmental degradation had already produced the operational effects cyber would have produced, and revealing

[20] For an accessible recent treatment, see Heather A. Conley and Matthew Melino, *America's Arctic Moment: Great Power Competition in the Arctic to 2050* (Center for Strategic and International Studies, March 2020). For the legal dimension of NSR governance, see Andreas Østhagen et al., works on Arctic maritime law.
[21] For the Congressional Research Service treatment, see CRS, "Changes in the Arctic: Background and Issues for Congress," updated 2024. For Government Accountability Office work, see GAO, "Climate Change: Information on Potential Economic Effects Could Help Guide Federal Efforts to Reduce Fiscal Exposure," GAO-17-720, September 2017.
[22] White House, *The National Security Strategy of the United States of America* (December 2025). For analytical treatment of the Western Hemisphere reframing and the "Trump Corollary," see Rebecca Pincus, "Protecting the North American Arctic and Beyond," Foreign Policy Research Institute, November 2025.

capability for redundant effect would be a strategic mistake. From the defender's perspective, this is itself a finding: environmental crises in Alaska are operationally similar to gray-zone attacks. The defender cannot easily distinguish whether degraded communications, fuel shortages, and personnel absenteeism are the product of nature or of adversary action. That ambiguity is itself an adversary advantage, and the operational architecture must be designed for an environment in which the two are not separable.[23]

The puzzle addressed is whether the United States has the analytical frameworks and the institutional architecture to operate in an environment in which environmental degradation, infrastructure fragility, social misunderstanding of needs, and strategic competition are the persistent and intertwined features of the operational landscape. The wargame's answer is no.

IV. FRAMEWORKS AND THEIR LIMITS

Two traditions dominate how Arctic homeland defense is currently assessed in the United States. The first is the strategic-capability tradition that emerged from Cold War strategic assessment and has been reinvigorated by the return-of-great-power-competition framing since 2017. The second is the urban-vulnerability tradition that emerged from disaster research and emergency management and is most influentially represented by the Hazards-of-Place model and its derivatives. Both traditions illustrate real and important aspects of the Arctic challenge. Both also share a limitation as neither adequately measures the variable that the wargame found decisive for whether the United States can sustain operations in its Arctic territory under stress.

The strategic-capability framework assesses national power through measurement of forces, platforms, and gross indicators of national strength. Composite Indicator of National Capability (CINC) scores, GDP-based assessments, and order-of-battle inventories are characteristic outputs of this tradition.[24] The tradition has produced sophisticated assessments of Russian and Chinese Arctic capabilities: the Russian icebreaker fleet, the Chinese investment in dual-use Arctic research infrastructure, the modernization of bases on the Russian Arctic coast and in the Northern Sea Route corridor. These assessments are valuable, but are also incomplete in a specific way. They measure what each state has rather than what each state can do with what it has under stress, and underexamine what they would do. The variable they do not develop is adaptive problem-solving capacity: the institutional ability to recombine existing resources, coordinate across jurisdictions, and improvise responses to novel conditions.

This limitation matters because adaptive capacity, not raw capability, is what determines outcomes in the kinds of crises the Arctic now presents. The wargame demonstrated this directly. The U.S. Coast Guard's authority structure (the captain-of-the-port's five-hat configuration, the Maritime Security (MARSEC)

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[23] Red Team after-action discussion, Day 2 afternoon, April 2026. The strategic logic of substituting environmental degradation for cyber action, and the explicit articulation that "you already have a lot of degradation that you didn't need to attack anything," is captured in the Day 2 afternoon transcript

[24] For the Composite Indicator of National Capability, see J. David Singer, Stuart Bremer, and John Stuckey, "Capability Distribution, Uncertainty, and Major Power War, 1820-1965," in *Peace, War, and Numbers*, ed. Bruce Russett (Sage, 1972), 19-48. The framework remains in use in international relations research through the Correlates of War project at the University of Michigan.

level mechanism, the Maritime Operational Threat Response (MOTR) coordination call, the Federal Maritime Security Coordinator (FMSC) role) allowed Coast Guard participants to coordinate responses to multiple simultaneous threats using authorities that exist on paper but had not previously been exercised against a compound environmental and gray-zone scenario.[25] The Alaska National Guard's (AKARNG) Civil Support Team (CST), with hazmat identification and rapid testing capabilities, exists for exactly the kind of ambiguous-threat scenario introduced when the red team paid a teenager to leave a backpack (eventually found to be harmless) near sensitive port infrastructure. But the CST is invisible to capability-based assessments because it is structured as a state asset under dual-status command rather than as a federal force. The State of Alaska's emergency response architecture, including the mutual aid system and the Emergency Management Assistance Compact integration with Lower 48 states, exists as a function of institutional relationships and rehearsed coordination that capability-based assessment cannot easily measure. The wargame's most consequential findings were focused on whether the institutional architecture can use the capability that exists under stress.

The urban-vulnerability tradition takes a different approach and runs into a different version of the same limitation. Kevin Borden and colleagues' 2007 study "Vulnerability of U.S. Cities to Environmental Hazards" is among the most influential application of the Hazards-of-Place model to American urban areas.[26] The study assessed 132 U.S. cities using three indices: social vulnerability (demographic and economic indicators of differential capacity to prepare for, respond to, and recover from hazards), built-environment vulnerability (infrastructure-density indicators), and hazard exposure (frequency and severity of hazard events). The study found that the most vulnerable cities are concentrated in the eastern half of the United States, with regional variation in which underlying correlates matter most: social vulnerability assuming greater importance in the South and Southwest, built-environment vulnerability assuming greater importance in regions with high infrastructure density. On its own analytical terms, urban-vulnerability framework treats Alaskan cities as comparatively resilient against the indices it measures.[27]

However, the framework is also incapable of capturing the vulnerability the wargame uncovered, and its conclusion that Alaska's urban centers are comparatively resilient is precisely what the wargame contradicts. The framework treats each city as a unit of analysis and assesses its vulnerability against other cities on a comparative national distribution. This works well when the question is which cities are most vulnerable to a given class of hazard event. It does not work when the question is what happens to the surrounding state if a particular city is taken offline.

Anchorage's vulnerability is not primarily about its internal social or built-environment metrics. It is about Anchorage's role as the irreplaceable hub for an entire state with extreme rural-remote dependency.

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[25] U.S. Coast Guard, Sector Western Alaska, captain-of-the-port authorities are codified in 33 CFR Part 6 and related authorities.

[26] Kevin A. Borden, Mathew C. Schmidtlein, Christopher T. Emrich, Walter W. Piegorsch, and Susan L. Cutter, "Vulnerability of U.S. Cities to Environmental Hazards," *Journal of Homeland Security and Emergency Management* 4, no. 2 (2007): Article 5.

[27] Borden et al., "Vulnerability of U.S. Cities to Environmental Hazards." The framework's findings, on its own terms, suggest that Alaskan urban centers are comparatively resilient against the indices the framework measures. The wargame contradicts this conclusion for the specific case of Arctic homeland defense, where the vulnerability is categorical rather than comparative and where the relevant unit of analysis is the strategic system rather than the population.

Anchorage's failure does not make Anchorage residents more vulnerable on a national index. It makes the entire State of Alaska functionally non-operational, and it cascades through U.S. Arctic homeland defense posture across the entire North American Arctic. The foundational premise of an Alaskan crisis is that there is no fall back or continuity of operations (COOP) if Anchorage is offline without airport flowthrough and reliable road networks; contrasting sharply with continuity capacities of the Lower 48 where logistics and response times can be managed.

This reflects how the Homeland Security and Emergency Management tradition has historically conceived of its object. The discipline emerged from civil defense and disaster response traditions that thought in terms of populations experiencing disasters, with the largest geographic unit of concern typically the city, the state, or the FEMA region. The analytical machinery the tradition developed (vulnerability indices, social vulnerability scoring, hazard exposure assessment) assumes that strategic stakes scale with the size of the population affected. A hazard event that affects more people matters more than one that affects fewer, all else equal. The premise is essentially democratic: it treats analytical concern as proportional to the number of human lives directly disrupted.

The premise holds for a wide range of cases the framework was built to address. Hurricane impact on coastal urban areas, tornado outbreaks in densely populated regions, flooding in major river basins: for these, the population-weighted approach captures what is most consequential. The premise fails where strategic stakes are decoupled from population size. Alaska is such a case. Its 740,000 residents are roughly 0.22 percent of the U.S. population. Its strategic significance, measured by the share of U.S. homeland defense, missile defense, NORAD execution, Pacific power projection, satellite ground station coverage, and global air cargo throughput that depends on Alaska infrastructure, is qualitatively larger. The Borden framework cannot represent this decoupling because its indices weight cases by population at risk. A 280,000-resident urban area in the Lower 48 with social vulnerability indices comparable to Anchorage would score similarly on the framework's measures. The strategic consequences of operational disruption in those two urban areas are not remotely comparable. The framework cannot register the difference because it was not designed to.

This is the deeper version of the critique. The framework's twenty-year-old conclusion that Alaskan urban centers are comparatively resilient is wrong not because it misidentifies the unit of analysis, but because it applies machinery that holistically discounts what is at stake in Arctic homeland defense. The same framework applied to Helsinki or Tallinn would be misleading for the same reasons. Finland and Estonia have small populations by NATO standards, roughly 5.5 million and 1.3 million, and would score modestly on population-weighted vulnerability indices. NATO planning does not treat them as modestly significant. Both are recognized as strategic anchors whose loss would substantially degrade alliance position, and the institutional and infrastructural investments NATO makes in their defense are proportional to that strategic significance rather than to their populations.[28] The United States has not extended the same logic to its own Arctic state.

[28] For the NATO Defence Planning Process, see NATO, "Defence Planning Process," available at https://www.nato.int/cps/en/natohq/topics_49202.htm.

The framework's age compounds the problem. Twenty years is not long in the history of social science, but it is long in the history of the threat environment the framework was built to inform.[29] The framework predates the smartphone, the supply-chain fragility that COVID exposed, the documented foreign information operations targeting American communities since 2016, and the cryospheric acceleration of the past two decades.[30] It was a substantial contribution in its moment. That moment has passed. The Hazards-of-Place tradition remains influential and continues to inform federal, state, and local hazard planning, but it has not been seriously refreshed against the threat environment that has since emerged.[31] The wargame provides an argument that this analytical infrastructure is overdue for refresh.

Both traditions miss the key variable of adaptive problem-solving capacity at the institutional level. The wargame demonstrated this capacity in operation, but also demonstrated its limits. The same institutional architecture that produced creative coordination on the multi-vessel collision response struggled to bridge the two game participant groups that self-formed into silos - one focused on HADR, the other on homeland defense.[32] The same captain-of-the-port authority that enabled rapid security posture adjustment ran into the limits of FAA temporary flight restriction conversion when counter-drone authority had to be coordinated across five categories of authorized actors with distinct training requirements. Adaptive capacity is real and consequential, but it is finite, and it is degraded by exactly the institutional habits (sequential rather than parallel processing, siloed rather than bridged coordination) that the existing frameworks reinforce.[33]

V. THE ARCTIC ENVIRONMENT AND THE NEW REVOLUTION IN MILITARY AFFAIRS

The wargame's findings cannot be fully assessed without understanding two changes in the operating environment that have accelerated dramatically over the past two decades and that are reshaping what "operating in the Arctic" actually requires. The first is cryospheric, the systematic transformation of the physical environment that has historically defined what is possible and impossible in the high latitudes.[34] The second is technological, the emergence of unmanned and autonomous systems and artificial

[29] For a representative recent assessment, see Andreas Raspotnik, "The Future Battlefield Is Melting: An Argument for Why the U.S. Must Adopt a More Proactive Arctic Strategy," The Arctic Institute - Center for Circumpolar Security Studies, December 3, 2024. See also Abbie Tingstad et al., *Report on the Arctic Capabilities of the U.S. Armed Forces*, RAND, 2023.
[30] Intergovernmental Panel on Climate Change, *Special Report on the Ocean and Cryosphere in a Changing Climate*, ed. H.-O. Pörtner et al. (Cambridge University Press, 2019). The sea ice decline statistic is from the report's summary for policymakers.
[31] Turetsky, Merritt R., et al., "Carbon release through abrupt permafrost thaw," *Nature Geoscience* 13 (2020): 138-143. The 20% abrupt-thaw figure has become standard in the cryosphere literature.
[32] USCG representative, after-action submission, April 2026. The "two siloish" characterization is hersI.
[33] Alaska National Guard military fellow representative, after-action submission. The full framing is that "information operations, command architecture, and logistics must be stood up simultaneously on Day 1: there is no sequential phase in which it is safe to defer any of them."
[34] For a recent treatment of Arctic infrastructure vulnerability and adaptation costs, see Congressional Research Service, "Changes in the Arctic: Background and Issues for Congress," updated 2024.

intelligence as decisive factors in military operations in any contested environment.[35] Both changes interact with the institutional and infrastructural vulnerabilities the wargame tested.[36] Both magnify the importance of the physical substrate on which adaptive capacity must operate.

The cryospheric changes are documented extensively in the scientific literature and have direct operational implications. Approximately 20% of the Arctic land surface is now classified as vulnerable to abrupt thaw.[37] This involves discrete events of rapid permafrost degradation that cause ground collapse, drainage changes, and infrastructure damage at scales not seen with gradual warming alone. For Alaska specifically, the operational implications are significant and immediate. The Trans-Alaska Pipeline System runs across approximately 420 miles of permafrost terrain. Roads, runways, fuel storage facilities, communications towers, and military installations across the state are built on or anchored in permafrost soils whose load-bearing capacity is degrading as those soils warm. The Alaska Department of Transportation's permafrost monitoring program documents accelerating maintenance costs across the state highway system. The U.S. Army Corps of Engineers' analyses of Arctic infrastructure project significant requirements for adaptation investment across military and civilian infrastructure if current trends continue. These costs are not theoretical projections of future risk. They are observable degradation in existing infrastructure that planning, budgeting, and operational architecture have not adequately absorbed.

This finding is invisible to the analytical frameworks that currently dominate both strategic and emergency-management assessment in the United States. Strategic capability assessments measure forces, platforms, and gross indicators of national power. Urban vulnerability assessments measure cities against each other on social, built-environment, and hazard-impact indices.[38] Neither framework adequately captures the kind of vulnerability the wargame revealed. Anchorage's vulnerability is not comparative in the sense those frameworks assess. It is categorical. Its failure does not make Anchorage residents more vulnerable than Houston residents on a national index; it makes the entire State of Alaska non-operational, with consequences that propagate through U.S. homeland defense for the entire North American continent.

Sea ice variability compounds the picture. The historical pattern of predictable seasonal ice cover that allowed Arctic operations to be planned around relatively stable navigation windows has given way to a pattern of high interannual variability with extreme events that are increasingly difficult to predict at operationally useful timescales. The Bering Strait, on which the wargame's gray-zone scenarios depended

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[35] For an accessible treatment of the unmanned-systems and AI evolution in contemporary military operations, with attention to the Ukraine war's lessons, see Andrew Krepinevich, *The Origins of Victory: How Disruptive Military Innovation Determines the Fates of Great Powers* (Yale University Press, 2023).

[36] Specifically, the test of unmanned systems showed that there were no takers. Novel technology offered and things like drones, robots, or generative software were not considered for possible solutions. No one wanted to use them, trusted them, or wanted to play with new systems in the face of traditional crises and challenges.

[37] Intergovernmental Panel on Climate Change, *Special Report on the Ocean and Cryosphere in a Changing Climate*, ed. H.-O. Pörtner et al. (Cambridge University Press, 2019). The sea ice decline statistic is from the report's summary for policymakers.

[38] Examples of the strategic-capability tradition's contemporary application to Arctic assessment include Heather Conley and Matthew Melino, *America's Arctic Moment: Great Power Competition in the Arctic to 2050* (Center for Strategic and International Studies, March 2020); the various Atlantic Council reports on Arctic posture; and the Department of War's *Arctic Strategy* documents under both the Biden and Trump administrations.

for plausibility, has seen substantial winter ice anomalies in recent years. The Northern Sea Route along the Russian Arctic coast is increasingly navigable in summer but remains hazardous, and the legal and operational regime governing it is itself contested. The Northwest Passage through Canadian Arctic waters presents related complications. None of these maritime environments is becoming reliably easier to operate in; all are becoming more variable, with extreme events that test institutional capacity to respond on shorter timescales than historical planning assumed.

Storm patterns have shifted as well. The 2022 storm that struck the western Alaska coast (Typhoon Merbok) produced coastal damage across Bering Sea villages that had historically been protected by seasonal sea ice. The damage prompted federal disaster declaration and a sustained mutual-aid response, including National Guard activation and inter-village resource sharing. The Merbok response is documented as one of the largest civilian evacuations in Alaska history at the village scale, and several wargame participants who had been involved in that response brought operational knowledge from it into the gameplay. The storm is a marker of a changing baseline rather than a one-time event. Communities that had not previously needed to plan for coastal storm surge now need to. Communities that had relied on seasonal sea ice as natural infrastructure now operate without it for longer periods of the year.

Wildfire patterns have shifted in the interior as well. Alaska's wildfire season has lengthened and intensified over the past two decades. The 2019 fire season burned approximately 2.6 million acres; the 2022 season burned approximately 3.1 million acres. Smoke from interior fires regularly disrupts commercial aviation across south-central Alaska during summer months. The 2021 season included sustained smoke events that grounded fixed-wing aviation across the Anchorage Bowl for periods of days.[39] The wargame's volcanic-ash scenario tested a more severe version of an aviation-disruption pattern that Alaska already experiences in less extreme forms during summer fire seasons. Participants drew on this experience directly during gameplay.

These cryospheric changes interact with the second change in the operating environment: the evolution of military operations toward unmanned, autonomous, and AI-enabled systems. The 2010s saw the consolidation of precision strike as the dominant paradigm of military advantage. The 2020s are seeing the emergence of a different paradigm in which mass-deployable unmanned systems, sensor networks, AI-enabled decision support, and distributed command and control are reshaping what operational advantage looks like in contested environments. The Ukraine war has accelerated this evolution and made the operational implications visible. The Russia-Ukraine conflict has demonstrated that inexpensive drones can impose disproportionate costs on expensive platforms, that civilian-grade communications and sensor technology can be weaponized at scale, and that the boundary between military and civilian operations is increasingly difficult to maintain in any contested environment.

For Arctic homeland defense, this evolution has specific implications that emerged from the wargame. The first is that the same environmental conditions that degrade U.S. capability (ash plumes, sea ice variability, storm severity, extreme cold) also degrade adversary capability. The wargame's gameplay demonstrated this directly when the Chinese Coast Guard vessel Nansha suffered a marine casualty in the ash plume and required U.S. rescue. The vulnerability is symmetric in important respects. The U.S.

[39] For Alaska wildfire trend data, see Alaska Department of Natural Resources, Division of Forestry & Fire Protection, annual season summaries, available at https://forestry.alaska.gov.

advantage, where it exists, is in adaptive capacity and rescue capability rather than in absolute platform survivability. The participant who articulated this during gameplay captured it: "they don't have operational performance in these conditions. We're going to have to go save their ass."[40]

The second implication is that unmanned systems do not fully substitute for crewed platforms in the Arctic operating environment. Quadcopter drones suffer rotor degradation from ash and other particulates and operational duration limits from extreme cold. Fixed-wing unmanned aircraft face the same engine-ingestion vulnerabilities that ground crewed fixed-wing aviation in ash events. The wargame's exploration of unmanned cargo delivery to rural communities underscored both the potential and the limits of the capability. Launch Alaska, the Anchorage-based clean-energy and aerospace technology accelerator that participated in the wargame, brought direct operational knowledge of unmanned cargo delivery pilots conducted in Alaska over the past several years, including the demonstration of beyond-visual-line-of-sight operations in coordination with the University of Alaska Fairbanks. The capability exists in early-deployment form but is not at the scale or maturity that would substitute for the logistics infrastructure the wargame demonstrated as irreplaceable.[41]

The third implication is that the counter-drone authority architecture that exists in U.S. law is not designed for the kind of compound threat environment that has been demonstrated in other operational theaters. Under current statute, five categories of actors are authorized to engage drones in U.S. airspace: certain Department of War personnel, certain DHS personnel, specifically designated law enforcement officers, certain critical-infrastructure-protection personnel, and tribal authorities with specific FBI-coordinated training under 14 CFR 124/130i. Engagement windows are typically sixty to ninety seconds from identification to intercept. The legal architecture assumes that the engaging actor has the right authorities, the right training, the right equipment, and the right relationship to the protected facility. In the wargame, none of these conditions held cleanly for the Port of Alaska. The conversation about who could legally take down a drone over the port (and under what authority, with what equipment, and with what coordination) showed an institutional gap that current planning frameworks do not address.

The interaction between cryospheric change and the new RMA produces a specific operational reality that the wargame illuminated. Environmental degradation in the Arctic increasingly produces the same operational effects that gray-zone activity produces in other theaters: degraded communications, supply-chain interruption, infrastructure failure, personnel attrition. From the defender's perspective, these effects are operationally indistinguishable. Whether a communications outage results from ash interference with satellite signals or from adversary action against undersea cable infrastructure may take days to determine, though the operational response must proceed regardless. The wargame's design exploited this fact directly. The volcanic ash event produced operational conditions within which gray-zone activity could occur in ways that the defender could not always identify as such in real time. The Red Team's strategic decision not to deploy cyber capabilities during the scenario reflected exactly this dynamic:

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[40] Statement by participant from the Alaska National Guard during Move 2 adjudication, Day 2 morning, April 2026. The dice roll on whether the Chinese cutter suffered a marine casualty came up 8 out of 12 (1-8 indicating breakdown), and the subsequent discussion is captured in the Day 2 morning transcript.

[41] For Launch Alaska's portfolio and Alaska unmanned systems pilots, see https://www.launchalaska.com. The University of Alaska Fairbanks Alaska Center for Unmanned Aircraft Systems Integration (ACUASI) holds FAA test-site designation and has conducted beyond-visual-line-of-sight cargo delivery demonstrations in coordination with industry partners.

environmental degradation had already produced the operational effects cyber action would have produced, and the marginal value of revealing capability was negative.

This convergence, environmental degradation as operationally similar to gray-zone action, and gray-zone action increasingly conducted under the cover of environmental degradation, is the deeper operating reality that the wargame's findings reflect. The Anchorage single-point-of-failure problem, the personnel-absenteeism finding, the natural-gas-turbine vulnerability, the international coordination deficits, each is a specific manifestation of a more general condition. Alaska sits at the intersection of accelerating environmental change and intensifying strategic competition, and the institutional architecture for managing either pressure in isolation does not adequately address the combination. The wargame is a test of what changes when both conditions are present simultaneously.

VI. WARGAME FINDINGS

The wargame produced findings across multiple operational domains and are organized around the central claim that Alaska has no scalable alternatives if its primary infrastructure fails.

A. Logistics: Anchorage as an irreplaceable hub.

Alaska depends on a small number of logistics nodes whose function cannot be substituted at scale by any combination of alternative infrastructure currently in place. The Port of Alaska is the only mainland Alaska port with gantry cranes and roll-on/roll-off ramps. Approximately ninety percent of consumable goods that enter Alaska transit through the Port of Alaska. Ted Stevens Anchorage International Airport handles cargo volumes that no other Alaska airport approaches. No military base in the Lower 48 states can provide the strategic range or Joint Base Elmendorf-Richardson or Eielson Air Force Base. Each node is irreplaceable in its specific function; the failure of any one cascades through the operations of the others.

Participants articulated this directly during gameplay. The participant from DHS CISA framed it most compactly: "Alaska is a spider web of single points of failure. The very center of that web is the Port of Alaska." The framing is operationally accurate. The combined capacity of Port Mackenzie, Whittier, Seward, Valdez, and Kodiak does not approach the Port of Alaska's throughput. None of these alternatives has gantry cranes or roll-on/roll-off capability comparable to the Port of Alaska's. Cargo that arrives in containers configured for Port of Alaska offload cannot be efficiently reconfigured for offload at alternative ports without infrastructure that does not currently exist. The participant from the Port of Alaska confirmed this directly during gameplay.[42]

The Port operates with three fuel terminals in normal configuration, with two operational at any given time and the third typically offline for maintenance. Move 2 inject included damage to one fuel terminal and a backpack incident near the valve operating system of another. Within forty-eight hours of gameplay

[42] Statement by participant from the Port of Alaska, Move 1, Day 1, April 2026. The technical specifications of Port of Alaska infrastructure are matters of public record; see Port of Alaska, "Port Modernization Program," at https://www.portofalaska.com.

time, both operational terminals were degraded, one through environmental damage, one through suspected sabotage. The Port's fuel offload capability fell to a single terminal at significantly reduced throughput. The Marathon refinery on the Kenai Peninsula was potentially impacted by ash. The Valdez export terminal was on the edge of the ash plume. The North Pole refineries were likely unaffected. The supply chain implication was direct: trucks running from North Pole and Valdez became the primary fuel distribution mechanism, with seasonal weight restrictions on the Alaska Highway as the limiting factor on throughput.

The lift capability the U.S. has is not configured for the kind of Arctic crisis the wargame modeled. The finding is not that the U.S. lacks lift capability in absolute terms. It is that the capabilities available do not meet the unique Alaskan needs. Naval Station Everett hosts guided-missile destroyers but not amphibious lift. The nearest amphibious assets are based at Naval Base San Diego. A two-week transit complicated by cold-weather operating limitations means that amphibious capability is not available within the relevant decision window for an Alaska crisis. The implication is that lift capability for sustained operations in Alaska depends on commercial maritime shipping, the Alaska Marine Highway System, and Air Force airlift through the C-17 and C-130 fleet. Each of these modalities has its own constraints. Commercial maritime is configured around Port of Alaska routing. The Alaska Marine Highway System is not designed for mass evacuation operations. Air Force airlift depends on receiving infrastructure and ramp space that would be degraded by ash. A third dimension of the logistics finding emerged through Move 1 gameplay and persisted through the remainder of the wargame. The Cook Inlet natural gas production infrastructure on which Anchorage's electrical generation depends is itself vulnerable to volcanic ash. Combined-cycle turbines are more sensitive to particulate ingestion than simple-cycle turbines; the combined-cycle units at the major south-central Alaska power generation sites would shut down under sustained ashfall conditions. The simple-cycle turbines at the Beluga power plant could continue operating with intermittent filter clearing, and hydroelectric generation through the Bradley Lake and Eklutna sources would continue, but the combined capacity would produce sustained brownout conditions across the Anchorage Bowl.[43] The participant from Launch Alaska articulated the cascading implication: without natural gas supply, there is no electricity in Anchorage; without electricity, the wastewater pump stations, fuel pump stations, and water treatment infrastructure progressively degrade.

This finding warrants specific methodological attention because the wargame designers had not anticipated it. The scenario was constructed to test air and maritime disruption. The natural-gas-turbine vulnerability emerged through participants applying operational knowledge during gameplay and through subsequent industry outreach during the white cell's adjudication process.[44] The finding is documented in the literature on volcanic-ash impact on gas turbines but had not been operationalized for Alaska planning purposes prior to the wargame. Alaska's energy planning should treat the natural-gas-turbine vulnerability as a primary planning variable rather than as an unmodeled assumption.

[43] For technical literature on volcanic ash impact on gas turbines, see, e.g., Casadevall, T.J., "The 1989-1990 eruption of Redoubt Volcano, Alaska: impacts on aircraft operations," *Journal of Volcanology and Geothermal Research* 62 (1994): 301-316. The literature is more developed for aviation than for ground-based power generation, but the underlying mechanisms (particulate ingestion into combustion stages) are common.

[44] Statement by participant from Launch Alaska, Move 1, Day 1, April 2026. The participant's outreach to industry contacts during white cell adjudication confirmed combined-cycle turbine vulnerability to ashfall and the relative tolerance of simple-cycle turbines at Beluga.

B. Personnel: direct labor, institutional consolidation, and the no-surge-support condition.

The wargame's personnel finding has three components that compound across the analysis. The first component is the absenteeism projection. The second component is the structure of who the absenteeism affects. The third component is the condition under which Lower 48 surge support that planning currently assumes would be unavailable or substantially constrained. Each component is operationally consequential and together they describe a personnel and workforce condition that current planning frameworks do not adequately address.

The absenteeism projection is thirty-five to forty percent among essential personnel during a sustained Alaska crisis. The figure emerged through participant discussion of the state Emergency Operations Plan, which explicitly assumes that many emergency responders will not report to work, and through participant operational experience with Alaska's recent disaster responses including the Cook Inlet earthquake and the response to Typhoon Merbok.[45] The figure is not a precise prediction; it is an operational planning assumption that the wargame validated through participant consensus drawing on professional experience. The driver of both components is family care rather than personal safety. Participants identified this directly during gameplay. The Alaska National Guard representative noted that when Mount Spurr erupted, his family was among the first to evacuate the Glenn Highway. The DLA Energy Quality Assurance Representative framed the institutional implication directly, drawing on the federal logistics community's experience during the COVID-19 pandemic period: family obligations come first in disasters, and the COVID period demonstrated that even the federal logistics infrastructure faces sustained workforce competition challenges during crisis. The Municipality of Anchorage Unified Command EOC Director, who runs Anchorage emergency operations as her actual professional role, articulated the operational implication: in disasters, family comes first; the question is whether the response architecture can accommodate this fact or whether it tries to ignore it. The implication for planning is that responder family support is an operational requirement that determines whether the response can function at all.[46]

The direct labor finding. The "first responder" category as it functions in American emergency management is a Lower 48 construct. It assumes a relatively elastic labor pool, geographic accessibility from neighboring jurisdictions, and a definable category of personnel (fire, EMS, police, public health, emergency management) whose absence in crisis is the principal personnel concern. The wargame's gameplay revealed repeatedly, without anyone naming it as a finding, that Alaska does not operate this way. The personnel who actually run the response in crisis are a much broader category whose roles are direct, physical, irreplaceable, and not substitutable from outside the state.

The longshoremen and stevedores who work the Port of Alaska are not first responders in the conventional sense. They are direct labor whose absence stops the port from functioning regardless of what other response architecture is in place. The work cannot be done remotely. It cannot be done by surge personnel flown in from the Lower 48 because the airport is degraded by the same crisis that is

[45] For the Merbok response documentation, see Federal Emergency Management Agency, Alaska Severe Storm, Flooding, and Landslides (DR-4672-AK), September 2022.

[46] For the operational significance of responder family welfare, see also the National Academies of Sciences, Engineering, and Medicine, *Healthy, Resilient, and Sustainable Communities After Disasters: Strategies, Opportunities, and Planning for Recovery* (Washington, D.C.: National Academies Press, 2015).

degrading the port. The labor is direct, physical, and place-bound. The wargame highlighted this when participants discussed the manual labor required to operate the gantry cranes and roll-on/roll-off ramps. The Port of Alaska's strategic seaport function depends on workforce that cannot be substituted, some of whom hold positions that are specialized trades that cannot be quickly replaced within the relevant decision window of an Arctic crisis.[47]

The deicing crews at Ted Stevens Anchorage International Airport and at Joint Base Elmendorf-Richardson are not first responders. They are direct labor whose absence stops aviation operations regardless of whether the aircraft are available, the runways are clear, and the air traffic controllers are at their stations. Deicing is a specific skill set with specific equipment that requires hands-on operation in subzero conditions. The work cannot be substituted from outside the state because the same crisis that creates the deicing requirement creates the aviation degradation that prevents external personnel from arriving.

The snowplow operators and road maintenance crews that keep the major ground lines of communication passable are not first responders. They are direct labor whose absence shuts down the only ground transportation route out of south-central Alaska. The Alaska Department of Transportation's snow removal capacity follows priorities, main arterials first, then secondary roads, then neighborhoods, with residents in some cases waiting through accumulating snowfall. The participant from the Municipality of Anchorage Office of Emergency Management named this directly during gameplay. There is no surge force for snow removal beyond the limited contractor base that already exists.[48] The Lower 48 mutual aid that exists for fire response and emergency medical services does not exist for snow removal in Alaska because no other state has the equipment, the operator skill, or the cold-weather operational capability at the relevant scale.

The fuel truck drivers who would have to run the Valdez-to-Anchorage and the North Pole-to-Anchorage routes if the Port of Alaska fuel terminal is degraded are not first responders. They are direct labor with commercial driver licenses, hazmat endorsements, and Alaska-specific winter driving experience. The seasonal weight restrictions on the Alaska Highway that the Alaska Department of Transportation can theoretically waive in crisis require the drivers themselves to be available to drive. The wargame showed this finding in the discussion of why the Defense Logistics Agency's fuel distribution plan depended on a small number of trucking companies, Shoreline, Weaver Brothers, Carlile, whose driver rosters are not interchangeable with Lower 48 trucking firms.

The natural gas turbine operators at Beluga, the wastewater treatment plant operators in Anchorage, the water treatment plant operators, the communications technicians who maintain the satellite ground stations and the undersea cable landings, the air traffic controllers at Ted Stevens and Fairbanks and the smaller field facilities, the harbor pilots who guide commercial vessels into the Port of Alaska through Cook Inlet's tidal complexity, the Alaska Marine Highway System crews, the bypass mail pilots who deliver the goods that sustain rural communities, the processing plant workers at the Marathon refinery on

[47] For Strategic Seaport designation, see U.S. Transportation Command, "Strategic Seaports," and Maritime Administration documentation. There are seventeen federally designated Strategic Seaports nationwide; the Port of Alaska is the only one in Alaska.

[48] Statement by participant from the Municipality of Anchorage Office of Emergency Management, Day 2 afternoon, April 2026. The snow removal limitation was discussed during the Move 3 wrap-up.

the Kenai and at the North Pole refineries, the construction equipment operators who would be required to clear ash from runways and from critical infrastructure, the hospital cleaning and food service workers without whom hospitals cannot function, none of these is a first responder in the conventional emergency management sense. All of them are direct labor whose absence stops a function the response architecture depends on. None of them is substitutable from outside the state within the relevant decision window of an Arctic crisis.

The thirty-five to forty percent absenteeism projection applies to all of these categories. The family-care driver that produces conventional responder absenteeism produces equal or greater absenteeism among direct labor categories that have less institutional accommodation for crisis-time family obligations. A Coast Guard sector commander has institutional support structures designed around the assumption that personnel will deploy in crisis. A longshoreman, a deicing operator, a snowplow driver, a fuel truck driver, a turbine operator does not have equivalent institutional support. The personnel are more vulnerable to crisis-induced absenteeism, not less.

The institutional consolidation finding. Alaska's small population produces a specific institutional pattern that the wargame consistently touches on. The participants in the room knew each other. They had worked together across multiple emergencies. They had each been in roles other than the role they currently occupied. The Coast Guard sector participant had worked previous Coast Guard assignments in different sectors. The DHS CISA participant had been a state emergency manager. The municipality emergency manager had been a regional emergency manager in a different jurisdiction. The institutional knowledge is held by a small community of people who rotate through roles and who collectively constitute the operational expertise the state has. This functions as a strategic vulnerability in most turnover situations and force restructuring, as well as in crises events that force composition shifts quickly.

This is a strength under most operating conditions. It produces the kind of cross-organizational coordination that the wargame's Day 2 gameplay demonstrated. It is also a vulnerability. The same thirty-five to forty percent absenteeism cuts into a smaller absolute number of people whose individual absence has correspondingly larger institutional consequences. The Lower 48 model of distinct workforce categories with distinct family obligations does not capture the Alaska pattern in which the same small group of people staffs multiple critical functions simultaneously.

The wargame opened this dynamic indirectly through participants' working knowledge of each other's roles, family situations, and cross-organizational obligations. A nurse who serves in the volunteer fire department. A wastewater operator whose spouse is a school nurse and whose adult child is an Alaska National Guard reservist. A longshoreman whose secondary role as a Civil Air Patrol pilot is itself critical to search and rescue operations. The pattern is widespread in Alaska. The same family is providing direct labor in multiple sectors. When the family responds to the crisis as a family, evacuating together, sheltering in place together, prioritizing the most vulnerable member, multiple sectors lose personnel simultaneously.

The implication for planning is that the absenteeism projection's operational consequence is larger in Alaska than in equivalently-sized Lower 48 jurisdictions because the institutional consolidation amplifies the per-individual consequence. A thirty-five percent absenteeism rate in a workforce where each

individual covers two or three critical functions produces a substantially larger operational disruption than the same percentage in a workforce where each individual covers one function. The Alaska planning environment requires recognition that the institutional consolidation is distinct from the Lower 48 baseline.

The no-surge-support condition. The wargame's gameplay rested on an unexamined assumption that Lower 48 surge support would be available within the timelines the response architecture currently uses. Federal Emergency Management Agency Region 10 personnel would arrive within twenty-four hours. Emergency Management Assistance Compact requests would flow the next morning. Cold-weather National Guard personnel from Minnesota, Michigan, and Wisconsin would be available to deploy. The Department of War's lift capacity from Joint Base Lewis-McChord and elsewhere would be configured for Alaska deployment. The Coast Guard's Pacific strike team and incident management team would deploy from lower latitudes within seventy-two hours.

The wargame's participants drew on direct experience to support these assumptions. The Cook Inlet earthquake response did proceed on this timeline. The Mount Spurr eruption response did proceed on this timeline. The Typhoon Merbok response did proceed on this timeline.[49] The professional experience that informed the gameplay was dominated by cases where Lower 48 surge support was available. The participants treated Lower 48 surge availability as a planning given because their professional experience supported the assumption.

The empirical record of recent years does not support the assumption as a planning given. The Federal Emergency Management Agency's annual disaster declaration rate has been climbing since the mid-2010s. The 2024 Atlantic hurricane season produced sustained federal response operations across the Gulf Coast and the Southeast. Western wildfire seasons produce sustained federal response operations across California, Oregon, Washington, Idaho, and Montana every summer and increasingly into fall and winter. Civil unrest deployments since 2020 have produced multiple National Guard activations across multiple states simultaneously. Immigration enforcement operations during 2024 and 2025 produced sustained National Guard deployments to Southern border states and generated counter-deployments by Northern states in response to federal immigration enforcement activity in their jurisdictions. The pattern is that the Lower 48 in any given crisis window is increasingly likely to have concurrent demands on the personnel, equipment, and institutional capacity that an Alaska crisis would require to surge.

A notable recent example of directional inversion in National Guard and military support relationships occurred in January 2026. Up to 1,500 active-duty soldiers from the U.S. Army's 11th Airborne Division, based in Alaska, were placed on prepare-to-deploy orders for potential support to federal operations in Minnesota amid civil unrest linked to large-scale immigration enforcement in the Minneapolis-St. Paul area. This situation inverted the directional flow assumed in many Alaska contingency plans: instead of Alaska receiving mutual aid from Lower 48 states under Emergency Management Assistance Compact (eMAC) arrangements during a local crisis, Alaska-based forces were positioned to support a continental U.S. contingency. Similar patterns have been evident in recent years, with the Alaska National Guard supporting missions in Washington, D.C. (May 2026), Eastern Europe (2025), the U.S. southern border

[49] Statement by representative from, Anchorage Office of Emergency Management, Day 1 afternoon, April 2026. Quoted with permission.

(2024), and the Middle East.[50] This case illustrates that mutual aid relationships and resource-sharing assumptions in U.S. emergency planning, while generally reliable, can be strained or reversed under conditions of concurrent national crises. Although the wargame did not explicitly model this inversion, the findings should be interpreted with this dynamic in mind as a realistic feature of the operating environment.

The implication is that Alaska's response architecture must be configured for a scenario in which Lower 48 surge support is unavailable or substantially constrained during the relevant decision window. The "no scalable alternatives" framing applies not just to physical infrastructure within Alaska. It applies also to the surge support from outside Alaska that current planning treats as available. The Alaska response architecture must be capable of functioning from its own standing resources during periods longer than current planning assumes, and may need to be capable of functioning from its own standing resources indefinitely in scenarios where Lower 48 concurrent demands prevent surge support entirely.

The three personnel components compound. The direct labor finding extends the responder category to the full place-bound skilled workforce. The institutional consolidation finding identifies that the workforce is smaller in absolute numbers than Lower 48 equivalents and that per-individual consequences of absenteeism are correspondingly larger. The no-surge-support condition identifies that the Lower 48 capacity planning currently assumes will substitute for absent Alaska workforce may itself be unavailable. The cumulative personnel finding is that Alaska's response architecture depends on a workforce that is broader than conventional responder categories, smaller in absolute numbers than Lower 48 equivalents, and not substitutable from outside the state through surge mechanisms that current planning treats as reliable.

Alaska's response architecture depends substantially on the operational knowledge of specific individuals rather than on documented procedures. The Joint Medical Emergency Plan Group meets monthly and has detailed protocols for patient movement; those protocols live in the heads of approximately a dozen healthcare leaders. The Port Readiness Committee operates under SSI clearances with limited written documentation accessible outside the committee membership. The Civil Support Team's specific operational capabilities are known to its commander and to a small number of state coordination contacts.[51] When thirty-five to forty percent of the direct labor and conventional responder workforce does not report, and when the institutional consolidation amplifies per-individual consequence, and when surge support is not reliably available, the institutional knowledge that operates Alaska's response architecture is the variable most at risk. The response architecture's resilience is not primarily infrastructural; it is human and relational. The cumulative personnel finding is that the human and relational resilience is itself at risk under conditions the wargame demonstrated as plausible.

[50] The deployment is cited here as documentary evidence of the directional inversion of mutual aid under concurrent crisis conditions; this research takes no position on the appropriateness of the deployment or the political context within which it occurred.

[51] The Civil Support Team is a National Guard asset under Title 32 with specialized hazmat and CBRN response capabilities. The relevance to ambiguous-threat scenarios was identified by the Alaska National Guard participant during gameplay.

C. Evacuation: no organized movement, one road out

The wargame stressed the reality that Anchorage has no plan to evacuate itself in any conventional sense, and that the absence is by design rather than oversight. The Municipality of Anchorage Unified Command EOC Director described an approach built on facilitated self-evacuation rather than organized convoys: traffic management would rely on traffic cones rather than law enforcement officers stationed at intersections, on the reasoning that officers tend to slow traffic rather than speed it.[52] The system assumes residents leave in their own vehicles, supported by infrastructure, with host-community coordination for the smaller number who cannot self-evacuate. This is a defensible plan for the hazard it was built around, however, it is not a plan that scales to a sustained, statewide event.

The constraints the discussion exposed are physical and unforgiving. Anchorage's roughly 280,000 residents are too many to move through any centrally directed evacuation using the extant civilian systems. The Seward Highway is the only ground route out of Southcentral Alaska. The Glenn Highway - the primary route to Fairbanks - is, in the words used during play, limited and treacherous, with winter weight restrictions that constrain throughput at exactly the season the wargame scenario modeled. If even 30% of the population sought some level of evacuation assistance, that is roughly 84,000 people, a number that pushes hard against Fairbanks' capacity to receive them and against the road system's capacity to carry them. There is no maritime or air alternative at that scale; the same ash plume that precipitates the evacuation grounds the aircraft and degrades the port.

Two human-systems findings sharpened the operational picture. The first is that disaster planning consistently assumes people behave rationally, and people behave humanely. The Hurricane Katrina precedent, raised during play, is the canonical example: residents refused to evacuate without their pets, which traumatized first responders and degraded cooperation. Alaska's plans now account for pets and service animals, but the deeper point generalizes: evacuation behavior is shaped by attachments and obligations that planning models tend to omit. The second finding is that the population-mobility dynamic compounds the personnel problem documented above. Roughly one-quarter to one-third of the affected population leaving or sheltering in place does not merely create an evacuation-management challenge; it removes the manual labor that runs the port, the airport, and the utilities, at the same moment those facilities are most needed.[53] Evacuation and workforce collapse are not separate problems. They are the same population dynamic viewed from two angles. There are no scalable alternatives. The single road, the single hub, and the single season combine so that a movement that would be merely difficult in a Lower 48 metropolitan area becomes, in Anchorage, a problem with no good answer under the conditions the wargame modeled.

D. International coordination: NORAD's structural gap.

The wargame's international coordination findings cluster around a single structural observation: the alliance architecture currently in place is not configured for the kind of Arctic crisis the wargame modeled, contributing to two oversights in the play: no obvious mechanism for engaging Canada’s

[52] Statement by representative of Anchorage Office of Emergency Management, Day 1 afternoon, April 2026. Quoted with permission.
[53] Alaska National Guard representative, after-action submission. Quoted with permission.

maritime elements for assistance; misalignment in bureaucratic levels for interactions (e.g., the city of Anchorage was not positioned to directly ask for assistance); and the assumption by players that someone else would be coordinating with allies. Part of this could have been selection bias: no USN officers played on blue and we did not model Department of State representation. However, improved inclusion of stakeholders would not have shifted that Anchorage officials may need to operate beyond municipal authority structures to solve specific resilience problems or take full advantage of possible assistance in a "fight tonight" framing. The most concrete instance was the deployment of HMCS Vancouver, a Halifax-class frigate with anti-submarine warfare capability, from Canadian Forces Base Esquimalt in response to the scenario. The deployment was made unilaterally by Canada as an inject rather than through game channels. The military participants in the wargame identified this immediately. NORAD does not have a standing maritime mission. The joint command is configured for counter-air-breathing threats. For a maritime crisis with strategic competition dimensions, even close allies must act on their own authority because the alliance structure does not accommodate the requirement.

This is a finding with direct policy implications for NORTHCOM, as well as PACOM, C3F and C7F with their respective Theater ASW stakeholders, treading into sensitive arenas with respect to readiness and capacity in respective Exclusive Economic Zones (EEZs). The U.S.-Canada alliance architecture in the Arctic has accumulated over decades and reflects the threat environment of the late twentieth century. Russian bomber incursions are a problem NORAD can address through its established authority. However, the list of contemporary Arctic threats that fall outside NORAD's standing mission is longer than the list of threats it does address. These include Russian icebreaker repositioning toward Nome and Kotzebue with humanitarian framing; the Chinese Coast Guard activity in the Aleutians; Russian submarine tampering with subsea cables; and wider alliance coordination challenges across respective services.

The wargame revealed a second international coordination finding about reactive engagement. Contact with Canadian authorities during the scenario was initiated only after critical subsea cable infrastructure was disrupted. Earlier engagement would have served U.S. operational interests but was not built into the response architecture. One solution is proactive bilateral engagement during peacetime to establish coordination mechanisms that do not depend on crisis-time activation. The Five Eyes (FVEY) information sharing architecture provides some of this for intelligence purposes but is not designed for operational coordination of the kind the wargame underscored as necessary.

The Chinese Coast Guard vessel Nansha, modeled in the scenario as a Zhaotou-class cutter approaching the Aleutians to render unilateral humanitarian assistance, suffered a marine casualty in the ash plume during Move 2 gameplay. The U.S. Coast Guard's strategic response, rendering aid to the Chinese vessel rather than allowing the Chinese to coordinate their own rescue, collapsed the Chinese humanitarian justification and produced an information-operations advantage for the United States. The relevance is twofold. First, environmental conditions degrade adversary capability the same way they degrade U.S. capability, and the adversary's reduced operational performance in Arctic conditions is a U.S. asset rather than a U.S. concern. Second, the U.S. institutional architecture must be able to recognize and exploit these opportunities in real time, which requires Coast Guard authorities that do not depend on State Department coordination cycles measured in days.

E. Information environment: clearance as chaos.

The wargame's most novel set of findings concerns the information environment. Three findings emerged that together describe a vulnerability the existing planning architecture does not address.

The first finding is something that can be characterized as "clearance as chaos." Personnel with security clearances in the wargame had access to information about adversary threat activity that uncleared personnel in the same room did not have. The Russian submarine in proximity to undersea cables at the time of the cable tampering was visible to cleared participants and invisible to uncleared participants. The wargame's gameplay revealed this gap within the room. Cleared officials know more about adversary threats to American communities than the American public knows, and the existing public affairs frameworks are not designed to bridge this gap. Traditional liaison and public information officer frameworks assume that the cleared/uncleared distinction does not generate operationally significant information asymmetries between government and public. The wargame demonstrates that this assumption is no longer valid.

The second finding concerns the FBI prioritization dynamic. Wargame participants subordinated their responses to FBI counterterrorism procedures during the backpack incident, even after the backpacks were determined not to contain explosives. The institutional hierarchy gives counterterrorism procedural primacy over humanitarian response, even when the actual threat is ambiguous and the humanitarian situation is acute. The wargame demonstrated that this hierarchy produces exactly the cost-imposition dynamic adversaries seek to exploit. Ambiguous low-cost actions force high-cost institutional responses regardless of whether the action turns out to be a real threat. The backpack incident's resolution as non-explosive did not unwind the security posture elevations, the MARSEC level changes, the force protection condition (FPCON) adjustments, or the personnel diversions that the initial response triggered. The cost was locked in at the moment of initial response, not in the moment of determination.

The third finding concerns the Red Team's deliberate exploitation of these dynamics. The Red Team chose not to deploy cyber capabilities during the scenario because environmental degradation had already produced the operational effects cyber would have produced. The Red Team's strategic framing during the hot wash captured the underlying logic: cost-imposition does not require any particular adversary action; it requires that the defender's institutional response architecture produce disproportionate costs in response to ambiguous threats. The Red Team's only requirement was to generate ambiguity. The defender's institutional architecture did the rest.

The Red Team also articulated a specific information-operations playbook targeting Alaska Native communities. The playbook drew explicitly on documented Russian information operations targeting minority communities in the United States since 2016, which produced documented effects on voting behavior, protest activity, and inter-community trust during the 2016-2020 period.[54] The playbook's application to Alaska Native communities exploits real historical and cultural ties (the Aleut-Russian

[54] For documentation of Russian information operations targeting Black communities in the United States, see Renee DiResta et al., "The Tactics & Tropes of the Internet Research Agency," New Knowledge report for the Senate Select Committee on Intelligence, 2018; and U.S. Senate Select Committee on Intelligence, "Report on Russian Active Measures Campaigns and Interference in the 2016 U.S. Election," Volume 2: "Russia's Use of Social Media with Additional Views," 2019.

historical connection, the Inupiat-Chukotka cultural relationship, the Russian Orthodox religious heritage in some Alaska Native communities) to seed doubt about U.S. government commitment to Alaska Native welfare. The Red Team's framing was direct: "you only need a couple, and you have a couple of villages, and especially in our more northern [communities] that are currently going to the UN panel because we as a government have not taken care of our contaminants. So you have disenfranchised citizens that may not create an opportunity."[55]

The wargame's gameplay tested this dynamic through structured adjudication. White cell accepted a red cell play of Russian icebreaker repositioning toward Nome and Kotzebue offering humanitarian aid to remote Alaskan communities based on cross-Strait tribal ties. The room debated whether the communities would accept the aid, with a key question being how well the previous summer's subsistence hunt had gone. The dice rolls adjudicated: subsistence hunt success rolled 9 of 12 (so the hunt was adequate); whether any community members boarded the Russian vessels rolled 3 of 12 (1-2 indicating yes, so none boarded); whether communities accepted aid offered without boarding was treated as effectively certain because the legal authority to prevent it does not exist beyond 12 nautical miles and federal refusal does not bind community-level acceptance.[56] The wargame's finding is that the planning assumption of uniform Alaska Native refusal of adversary aid is wrong. Some communities, in some conditions, would accept aid offered with sufficient cultural framing. The finding does not establish which communities or which conditions; it establishes that the uniform-refusal assumption is unsafe.

F. Tribal and cultural dimensions.

The differential community response finding warrants additional attention because it intersects with the wargame's most significant methodological limitation. Alaska Native and tribal representation were not present in the room. The findings about tribal community response were produced by non-tribal participants reasoning about tribal behavior using operational knowledge from prior engagement with Alaska Native communities. The findings are consistent with historical patterns and with the dynamics the Red Team identified. They are not equivalent to findings that direct tribal participation would have produced. With that limitation explicitly stated, the wargame's findings on tribal and cultural dimensions are operationally significant.

Any assumption that Alaska Native communities are a single block with uniform interests is wrong; this was not a position held by all participants but the tribes were initially discussed as more of a collective block vice discrete entities. Alaska contains approximately 229 federally recognized tribes, organized through twelve regional Alaska Native corporations and additional village corporations established under the Alaska Native Claims Settlement Act. The communities differ substantially by language family (Yup'ik, Inupiat, Aleut/Unangan, Alutiiq, Tlingit, Haida, Tsimshian, Eyak, and the various Athabascan groups), by geography, by historical experience of contact with non-Native populations, by economic structure, and by religious heritage. The Russian Orthodox influence in Aleut, Alutiiq, and parts of the Yup'ik communities is historically significant and culturally durable. The Inupiat communities of the North Slope have different historical and cultural patterns. The Tlingit and Haida communities of

[55] Red Team participant statement during Day 2 afternoon hot wash. Quoted with permission under Red Team non-attribution agreement; the speaker is identified as a Red Team participant without further attribution.
[56] Day 2 morning transcript, Russian icebreaker scenario adjudication. The full dice-roll sequence and discussion is captured in the transcript.

southeast Alaska have different patterns again.[57] Geographic heterogeneity further complicates the operational environment. Alaska's expansive and culturally distinct communities are distributed throughout the state; consequently, planning that treats these communities as interchangeable is unlikely to survive operational contact during a catastrophic disruption.

The wargame opened a specific operational implication for the U.S. Coast Guard and other federal responders. Aid offered by adversaries outside U.S. territorial waters (beyond 12 nautical miles) cannot be legally prevented by the U.S. government. Community-level acceptance of such aid is similarly outside the U.S. government's prevention authority. The federal response options are limited to ensuring community safety during aid acceptance (preventing Russian or Chinese personnel from coming ashore, maintaining security perimeters during transfer operations) and to providing alternative aid that makes adversary aid less attractive.[58] The wargame's gameplay highlighted this directly: "if they want to walk out to the ice and accept the aid, I don't care if they do that. We probably wouldn't stop them. But we would facilitate safety and security of the stuff and make sure no Russians come with them."

Direct tribal engagement during peacetime is the only mechanism that adequately addresses these dynamics. Crisis-time engagement is too late. The relationships, trust, and cultural understanding required for effective response cannot be improvised under crisis conditions; they must be built during the long stretches of time when no crisis is occurring.

G. Institutional tempo: sequential versus parallel.

The wargame showed a finding about institutional tempo that emerged most clearly in the after-action submissions of two military participants. The Alaska National Guard representative articulated the framing directly: "The most consequential learning from our decision-making process was the cost of sequential rather than parallel planning. The DSC activation, the comms issues, aviation optimism assumptions, and the Alaska Marine Highway retasking were all identified early and executed late. Each delay in a real situation would have compounded; within the game Red was not allowed to keep moving so as to not overweight the asymmetry of action but real life situations would differ. The wargame demonstrated clearly that information operations, command architecture, and logistics must be stood up simultaneously on Day 1 as there is no sequential phase in which it is safe to defer any of them."[59]

This finding has structural depth that warrants explicit attention. The U.S. emergency response architecture defaults to sequential prioritization through the Incident Command System, the National Response Framework, and the various sector-specific protocols. Establish command first, then assess capability, then execute mission. The logic of sequential prioritization is sound under conditions where the threat is discrete, the situation can be stabilized, and the relevant authorities can be coordinated in a logical sequence. Under conditions of compound stress, persistent adversary activity, and degraded

[57] For an overview of Alaska Native tribal organization and the Alaska Native Claims Settlement Act, see Stephen Haycox, *Alaska: An American Colony*, 2nd ed. (Seattle: University of Washington Press, 2020). For the contemporary tribal landscape, see Bureau of Indian Affairs, "Indian Entities Recognized by and Eligible To Receive Services From the United States Bureau of Indian Affairs," *Federal Register* (annual publication).
[58] Day 2 morning transcript, Chinese cutter marine casualty adjudication. The dice roll came up 8 of 12 (1-8 indicating breakdown). The subsequent discussion of U.S. rescue response and its strategic implications is captured in the transcript.
[59] Alaska National Guard representative, after-action submission. Quoted with permission

communications, sequential prioritization produces exactly the windows that adversaries exploit. Adversaries operate in parallel because they do not have the institutional habit of sequential prioritization. Defenders default to sequence because that is how the institutional architecture is designed.

USCG representative's after-action submission identified the spatial-institutional version of the same finding. The wargame produced "two siloish" groups, one focused on HADR and one focused on homeland defense. The siloing was both a feature of the scenario design (which separated the two threads to allow each to be developed) and a realistic representation of how institutions actually behave under stress. Institutional response architecture defaults to organization-by-organization, problem-by-problem processing, and adversaries operate in the gaps.

The finding has direct policy implications. The standard joint planning approach assumes that command structures can be established in a logical sequence with adequate time for each phase. Future Arctic planning must assume that command structures, information operations, logistics, and tribal engagement must all be stood up simultaneously on Day 1, and optimally cultivated as part of Phase 0 for any Alaska operation. Pre-delegated authorities, pre-positioned supplies, pre-established relationships, and pre-trained personnel are prerequisites for the response architecture's basic functioning. Arctic infrastructure investment is a homeland defense problem that extreme weather is accelerating.

VII. ANALYSIS

The wargame's findings, taken together, support an analytical framework that differs from both the strategic-capability tradition and the urban-vulnerability tradition in several specific respects. Adaptive problem-solving capacity operates as the decisive variable. The analysis applies the adaptive-problem-solving framework developed in the author's separate doctoral research to the case of Arctic homeland defense. The framework's central claim is that the capacity of states to solve operational problems is the forgotten core of state power, and that capability-based assessments consistently do not consider this variable. The argument is developed at length in the dissertation and is not produced by this report.[60]

The wargame's findings, read through the framework's lens, establish that adaptive problem-solving capacity is what determines whether the State of Alaska can sustain operations under stress and, by extension, whether the United States can sustain its strategic posture in the North American Arctic. The Coast Guard's captain-of-the-port authorities, the Civil Support Team's hazmat identification capabilities, the cross-trained personnel who routinely loan themselves across organizations during emergencies, the mutual aid system that integrates Alaska's response architecture with Lower 48 states through the Emergency Management Assistance Compact, the institutional knowledge that lives in specific individuals, these are the elements that produced functional response in the wargame's gameplay. None of

[60] The analytical framework applied draws on the author's separate doctoral research, currently in preparation for her dissertation at the University of Chicago Department of Political Science. See Iverson (forthcoming 2026), "You Had One Job: Solving Problems as the Forgotten Core of State Power." The framework's central arguments are that adaptive problem-solving capacity is the forgotten core of state power and that capability-based assessments discount this variable and are developed at length in that work. This research project applies the framework to the empirical case of Arctic homeland defense rather than developing the framework itself.

these elements would appear in a capability-based assessment of U.S. Arctic posture. None of them would appear in a population-weighted vulnerability assessment of Alaska's urban centers. All of them were decisive for whether the wargame's response architecture worked.

Capability that exists on paper does not equal capability that can be applied in time for the relevant decision window. The Anchorage response architecture's adaptive capacity could not compensate for the loss of Cook Inlet natural gas supply because the loss propagated through electrical generation, water treatment, fuel pumping, and habitable temperature maintenance simultaneously. The institutional knowledge and coordination capacity that enables adaptive response cannot operate without electricity, communications, and habitable working conditions for the responders. Physical infrastructure is the substrate on which adaptive problem-solving operates, and the substrate's degradation degrades the capacity regardless of the institutional architecture's quality.

The framework claims that adaptive problem-solving capacity, physical infrastructure, and workforce are interdependent variables that must be developed jointly rather than optimized separately, a claim applied to the Arctic case. The wargame's findings support all three components. The recommendation for two deep-water ports and two airports represents an investment in the physical infrastructure necessary for adaptive problem-solving, especially when existing infrastructure degrades. This same logic extends to distributed energy generation, hardened communications, and workforce protection. Each of these constitutes a physical or institutional investment designed to prevent adaptive problem-solving capacity from collapsing under stress.

The contrast with the standard strategic-capability tradition is instructive. The strategic-capability tradition measures force structure, platform inventory, and gross indicators of national power on the implicit assumption that capability can be applied wherever it is needed. The Arctic operating environment does not satisfy this assumption. Fifth-generation aircraft evacuate to the lower forty-eight under sustained ash conditions; amphibious lift takes roughly two-weeks to transit from San Diego with cold-weather operating limitations; Naval Station Everett does not host the lift configuration required. Capability that exists on paper does not equal capability that can be applied in time for the relevant decision window. The variable that matters is not whether the United States has the capability but whether the institutional architecture can apply the capability under specific operating conditions. Adaptive capacity is a function of three interdependent components: physical infrastructure, institutional architecture, and the workforce that operates both.

The workforce component consists of the conventional responder workforce and the direct labor workforce that operates the physical infrastructure on which response depends. The two categories together are larger than conventional first-responder frameworks acknowledge. The workforce is smaller in absolute numbers in Alaska than in equivalently-strategic Lower 48 jurisdictions because Alaska's institutional consolidation produces high per-individual operational consequence. The workforce is not substitutable from outside the state through surge mechanisms that current planning treats as reliable because the Lower 48 in any given crisis window is increasingly likely to have concurrent demands on the same personnel, equipment, and institutional capacity.

The no-surge-support condition is a feature of the operating environment within which the adaptive capacity argument must be developed. The condition is the emerging pattern of American homeland

response in a period when the Lower 48 has its own substantial concurrent demands across natural disaster, civil unrest, immigration enforcement, and other categories of institutional commitment. The condition's character means that Alaska's adaptive capacity must be developed to function independently of Lower 48 surge support across periods longer than current planning assumes and, in some scenarios, indefinitely.

Arctic homeland defense requires investment in adaptive capacity across physical, institutional, and workforce dimensions simultaneously, configured for an operating environment in which Lower 48 surge support is not reliably available. The cumulative investment is substantially larger than current planning treats as required because current planning rests on the Lower 48 surge availability assumption that the wargame did not test and that the empirical record does not support. Strategic significance and population size are decoupled in ways that current frameworks cannot represent, and that this decoupling is itself a substantive finding of the exercise. The Borden framework's conclusion that Alaskan urban centers are comparatively resilient is produced by analysis that weights cases by population at risk. The framework is correct on its own terms. Anchorage *is* comparatively resilient if the question is "how vulnerable is this city's population to disaster events relative to other U.S. city populations." Anchorage is not resilient if the question is "what happens to U.S. strategic posture if this city's infrastructure becomes non-operational."[61]

The two questions produce different answers because they assess different objects. The first question assesses populations as the unit of moral and analytical concern. The second assesses strategic systems in which populations are embedded. The Homeland Security and Emergency Management tradition has historically prioritized the first question and underdeveloped the assessment criterion for the second. This is appropriate when the two questions produce similar answers and when strategic stakes scale roughly with population size. However, it is inadequate when the two questions produce different answers, as in the Alaska case.

Homeland defense planning in regions where strategic significance is decoupled from population requires different tools than homeland defense planning in regions where it is not. The tools needed for Alaska are tools that can represent the strategic-system dependencies that produce decoupling. Single-point-of-failure analysis for statewide systems. Cascade analysis across military, civilian, and infrastructure dependencies. Adversary-action assessment that captures the institutional vulnerabilities the wargame highlighted. None of this assessment infrastructure is standard in the existing emergency management discipline.

The Florida hurricane comparison is illustrative. Florida experiences sustained hurricane impact on a near-annual basis. The economic and humanitarian consequences are substantial; the National Guard activations, FEMA deployments, and federal disaster declarations follow established patterns.[62] The strategic consequences for U.S. posture globally are essentially zero. The same is not true for a

[61] USCG representative, after-action submission. Quoted with permission.

[62] For the FEMA National Risk Index methodology, see Federal Emergency Management Agency, "National Risk Index Technical Documentation," updated 2024, available at https://www.fema.gov/flood-maps/products-tools/national-risk-index. The Expected Annual Loss calculation methodology is documented in the technical materials.

comparable-scale operational disruption in Alaska. The reason is not that Alaska experiences worse weather or that its populations are more vulnerable in any social-vulnerability index sense. The reason is that what is physically located in Alaska, the missile defense field, the early warning radars, the Pacific air power forward presence, the strategic seaport, the trans-Pacific air cargo node, produces strategic consequences that follow from infrastructure degradation regardless of the population-level humanitarian dimension.

Investments in Alaska's physical and institutional infrastructure should be weighed against other strategic investments such as missile defense modernization, alliance commitments, Pacific posture on appropriate strategic-significance criteria, rather than on a population-weighted distribution against other regional investments.The current framework systematically misclassifies them and therefore systematically underweights them.

Operational practice meeting structural reality: a note on evacuation planning.

The evacuation finding identified that Anchorage's evacuation planning relies on traffic management through cones and barriers rather than law enforcement deployment, on the operational logic that police presence slows evacuating traffic rather than facilitating it. The Municipality of Anchorage Unified Command EOC Director articulated this approach from her position as the operational practitioner who actually runs Anchorage emergency management. Her professional knowledge is direct rather than imported from Lower 48 frameworks, and the traffic-cones approach reflects sound operational practice within the existing planning architecture.[63]

The wargame co-lead, during the post-game synthesis, raised a productive reservation. Even sound traffic management practice faces conditions in Alaska's road infrastructure that may exceed its design parameters. The Alaska Highway is the only ground route out of south-central Alaska to the lower forty-eight. Its winter conditions are treacherous; its refueling infrastructure is limited; its capacity is constrained by seasonal weight restrictions and by the realities of winter driving on a two-lane highway across multiple thousand miles of remote terrain. Even in summer, the evacuation of population at scale from south-central Alaska through the Alcan would face physical capacity limits that traffic management cannot eliminate. A major pile-up at the Tok Cutoff, at the Yukon border crossing, or at any of the constrained points along the route would compound rapidly into operational paralysis regardless of how well traffic cones direct flow at the origin. The engagement between these two perspectives undergirds the productive tension between operational best practice and structural reality. The Anchorage emergency management lead is correct that traffic cones produce better evacuation outcomes than law enforcement deployment within Anchorage's road infrastructure capacity. The wargame co-lead is correct that the broader Alaska road infrastructure has capacity limits that the most efficient traffic management within Anchorage cannot exceed. Both perspectives together produce the report's conclusion that the evacuation architecture's deficiencies are not primarily about traffic management practice but about physical infrastructure capacity. This highlights the possible advantages in building out more robust Noncombatant

[63] Statement by participant from the Municipality of Anchorage Office of Emergency Management, Move 1, Day 1, April 2026. The traffic-management logic ("we would not be using police officers...in an evacuation situation, when people see law enforcement officers, it slows everything down") is captured in the Day 1 afternoon transcript.

Evacuation Operations plans (NEO), potentially in tandem with private sector stakeholders such as cruise lines, which were thought experiments raised during play that currently have no through-thread.

How existing frameworks generate Alaska's mis-classification, and what an alternative measurement approach requires.

The argument that strategic significance and population size are decoupled in the Alaska case rests on a methodological claim that warrants direct engagement with the analytical frameworks currently in use.

The Hazards-of-Place tradition and its derivatives.

The Hazard of Place tradition constructs its assessment through three composite indices. The Social Vulnerability Index aggregates approximately a dozen demographic and economic variables, population density, racial and ethnic composition, age distribution, gender ratio, housing tenure, income, education, employment status, and similar variables, into a single score intended to capture differential capacity to prepare for, respond to, and recover from hazard events. The Built Environment and Infrastructure Vulnerability Index aggregates infrastructure density variables, housing units per square mile, commercial establishments, road density, manufacturing concentration, into a single score intended to capture exposure of built assets to hazard impact. The Hazards Index aggregates frequency and severity data for specific hazard types, primarily hurricanes, tornadoes, floods, earthquakes, and wildfire, weighted by area exposure. The three indices combine into a composite vulnerability score against which the framework's 132-city sample is ranked.[64]

The framework's machinery generates Anchorage's relative-resilience classification through specific mechanisms that warrant examination. Anchorage scores low on built-environment density because the indicators the framework uses, housing units per square mile, commercial establishments per square mile, road density, reflect Lower 48 patterns that Alaska's geography does not produce. Anchorage's housing stock is dispersed across a metropolitan area whose density is low by national standards. The framework treats this low density as low built-environment vulnerability. The framework's logic is that dispersed infrastructure is less concentrated and therefore less vulnerable to acute impact. The logic is sound for the cases the framework was designed to assess. It is wrong for the Alaska case because Anchorage's low density coexists with concentrated function. The Port of Alaska is a single facility. Ted Stevens is a single facility. JBER's strategic infrastructure is concentrated in a small number of buildings. The framework's density measures do not capture functional concentration. The framework's structure systematically mis-categorizes Anchorage as low-vulnerability because Anchorage's vulnerability is functional rather than density-based.

Anchorage scores moderately on social vulnerability because its demographic profile does not match the indicators the framework treats as elevating risk. Anchorage's age distribution, income levels, housing tenure patterns, and educational attainment are not extreme by national standards. The framework's social vulnerability indicators were developed to capture differential capacity within American urban areas of broadly similar economic profile. Anchorage's social vulnerability scores in the moderate range. The framework concludes that the Anchorage population is moderately equipped to respond to disaster events.

[64] Most prominently the Hazards-of-Place tradition associated with Susan Cutter and her collaborators

The conclusion is correct on the framework's terms for the question the framework is designed to answer: relative differential capacity within the U.S. urban distribution. The conclusion is not a measurement of whether the response capacity Anchorage has would be sufficient under the specific operating conditions the wargame revealed as the framework cannot represent the gap.

Anchorage scores low on hazard exposure because the framework's hazard-frequency weighting privileges hurricane, tornado, flood, earthquake, and wildfire events for which substantial historical data exists. Volcanic ash events, sustained extreme cold compounding with infrastructure failure, gray-zone activity from foreign actors, and the cascading multi-hazard scenarios the wargame modeled either do not appear in the framework's hazard taxonomy at all or appear with low frequency weightings. The framework does not have a category for compounding environmental degradation in the presence of strategic competition. The framework's conclusion that Anchorage faces low hazard exposure follows from the framework's hazard taxonomy, where the taxonomy is the limit. The framework cannot register hazards it does not categorize. The cumulative effect is that the Borden et al. framework produces a mis-classification of Anchorage. It reveals in this case that the framework is operating as designed against a case the framework was not designed to assess. The framework was developed to compare American cities on differential vulnerability to disaster events within the assumptions of mid-2000s emergency management. Alaska did not fit those assumptions in 2007 and fits them less well now.

The FEMA National Risk Index, currently in active use as the federal government's principal community-level risk assessment tool, has a similar architectural problem. The Index calculates Expected Annual Loss for each U.S. census tract by combining hazard exposure data, social vulnerability indicators, and community resilience indicators into a single risk score. The Expected Annual Loss calculation weights cases by population at risk and by historical loss data for the relevant hazard types. Alaska's small population produces low Expected Annual Loss values for most Alaska census tracts because the population-weighting reduces the absolute loss estimates. The limited historical loss data for Alaska-specific hazard types such as volcanic events, sea ice variability impacts, permafrost-related infrastructure failure produces low hazard exposure values because the framework's historical-frequency weighting cannot accommodate events that occur infrequently but with high consequence.[65] The National Risk Index therefore concludes that most Alaska census tracts are at relatively low overall risk. The conclusion follows logically from the framework's structure. The conclusion is wrong for the same reasons the Borden conclusion is wrong. The framework's structure assumes that risk scales with population at risk and with historical hazard frequency. Neither assumption holds for the Alaska case.[66]

The Department of Homeland Security's National Infrastructure Protection Plan (NIPP) and its sector-specific implementations represent a different analytical tradition. The NIPP identifies Critical Infrastructure and Key Resources across sixteen designated sectors and maintains decentralized

[65] Intergovernmental Panel on Climate Change, *Special Report on the Ocean and Cryosphere in a Changing Climate*, ed. H.-O. Pörtner et al. (Cambridge University Press, 2019). The sea ice decline statistic is from the report's summary for policymakers. The 30-50% circumpolar infrastructure risk projection is widely cited in the cryosphere literature; see also Hjort, Jan, et al., "Degrading permafrost puts Arctic infrastructure at risk by mid-century," *Nature Communications* 9 (2018): 5147

[66] Borden et al., "Vulnerability of U.S. Cities to Environmental Hazards." The methodological detail in this paragraph is drawn from the framework's documentation.

identification processes through sector-specific agencies.[67] The framework produces classifications of specific facilities by their criticality to sector function. The Port of Alaska is identified in the Transportation Systems sector lists as a Strategic Seaport. Ted Stevens is identified in the Transportation Systems sector lists as a major air cargo facility. JBER, Eielson, Fort Wainwright, and Fort Greely are identified in the Defense Industrial Base sector lists as DoW-essential facilities. Clear Space Force Station is identified in the Government Facilities sector lists. The Trans-Alaska Pipeline System is identified in the Energy sector lists.[68]

Although the infrastructure is identified, the framework does what it is designed to do at the level of individual asset classification. What the NIPP and its derivatives do not do is assess cumulative dependency of multiple critical infrastructure functions on a single geographic node. The framework's approach is sector-specific by design. Inter-sector dependency analysis is not a standard NIPP output. The cumulative effect of degrading the Port of Alaska, Ted Stevens, JBER, the Cook Inlet energy infrastructure, and the Anchorage communications nodes simultaneously is not represented in the framework's analysis. As such, while the framework can identify each as critical, it cannot represent that they are all located in the same urban area whose simultaneous degradation would cascade through every other critical-infrastructure category in the state. The wargame's central finding is precisely about this inter-sector cascading dependency, which the NIPP cannot represent because it was not designed to. This is not a critique of the NIPP as a framework. The NIPP does sector-specific identification well and has produced substantial improvements in U.S. critical infrastructure protection at the sector level over the past two decades. The critique is that no framework currently in standard federal use does what the NIPP does not do, inter-sector cascading dependency analysis at the geographic-node level. The gap is foundational to the framework architecture, and the wargame demonstrates that the gap operationally matters.

The strategic-capability tradition and its limits.

The strategic-capability tradition has a different but related problem. The Composite Indicator of National Capability (CINC) score, developed in the 1960s and still used in international relations research, aggregates six variables, military expenditure, military personnel, energy consumption, iron and steel production, total population, and urban population, into a single national capability score. The framework was developed for inter-state comparison and produces useful aggregate assessments of relative national power. CINC scores enable comparative analysis of how national capabilities have evolved over time and across cases. The framework does not have a mechanism for assessing where within a national territory the capability is physically located, what infrastructure dependencies the capability requires to function, or how the capability degrades under specific operating conditions.[69]

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[67] For the National Infrastructure Protection Plan, see Department of Homeland Security, "NIPP 2013: Partnering for Critical Infrastructure Security and Resilience," and the sixteen sector-specific implementations. The Strategic Seaport designation is administered by the U.S. Transportation Command and the Maritime Administration.
[68] For Arctic renewable energy research and deployment, see University of Alaska Fairbanks, Alaska Center for Energy and Power, "Arctic Energy Office" publications, available at https://acep.uaf.edu. For Alaska Industrial Development and Export Authority programmatic infrastructure, see https://www.aidea.org.

[69] For the Composite Indicator of National Capability, see J. David Singer, Stuart Bremer, and John Stuckey, "Capability Distribution, Uncertainty, and Major Power War, 1820-1965," in *Peace, War, and Numbers*, ed. Bruce

RAND's Arctic capability assessments, the Atlantic Council's Arctic posture analyses, and similar work apply more granular analysis to specific platforms and missions.[70] The Department of War's Arctic Strategy documents and the Pentagon's regional combatant command posture statements provide more operationally specific assessments. Most of this work shares an assumption inherited from the strategic-capability tradition: that capability can be applied wherever it is needed, with the principal question being whether the United States has the capability and whether that capability is sufficient relative to adversary capability. **The wargame demonstrated this assumption to be false for the Arctic case.** The capability that exists in Alaska is not equivalent to capability that can be applied in Alaska under stress. Fifth-generation aircraft evacuate to the Lower 48 under sustained ash conditions; their capability exists on paper for the duration of the crisis but is not available for application in theater. Capability that exists outside Alaska is not equivalent to capability that can be applied in Alaska in time for the relevant decision window. Amphibious lift, CVNs, or logistic support ships from San Diego require two-weeks of transit with cold-weather operating limitations; its capability exists but cannot be applied within the operational tempo the crisis requires. Contracting commercial shipping is one possible alternative, but given how ad hoc this would be to execute in a sudden crisis, it may be even slower to bureaucratically architect than using military assets. The strategic-capability tradition's frameworks do not have mechanisms for representing time-to-application or operating-condition-dependent capability degradation. The frameworks therefore produce assessments of U.S. Arctic capability that systematically overestimates what is actually available for application under crisis conditions.[71]

The Arctic Capabilities study RAND produced for the Department of War in 2023 is a substantial contribution that engages many of these issues directly and makes specific recommendations about capability investment. The study's methodology is more sophisticated than the bare CINC approach and incorporates operational-condition assessments that the older strategic-capability tradition does not. The study is closer to the kind of framework the Arctic case requires. It nevertheless inherits the assumption that the question is primarily about capability quantity rather than about institutional architecture and adaptive capacity. The wargame's findings suggest that the Arctic case requires further development of the framework even beyond what the RAND study advances.

The NATO alternative: how strategic-significance-weighted assessment works in practice.

NATO's planning frameworks provide an operational alternative that warrants direct examination because they produce assessments of small but strategically significant states that population-weighted approaches would not produce. The NATO Defence Planning Process operates through a four-year cycle in which member states submit National Plans against alliance-wide Minimum Capability Requirements.[72] The Minimum Capability Requirements are not derived from member-state populations. They are derived from alliance strategic requirements, what the alliance needs to be able to do collectively, distributed

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Russett (Sage, 1972), 19-48. The framework remains in use in international relations research through the Correlates of War project at the University of Michigan.

[70] Abbie Tingstad et al., *Report on the Arctic Capabilities of the U.S. Armed Forces*, RAND, 2023.

[71] Examples of the strategic-capability tradition's contemporary application to Arctic assessment include Heather Conley and Matthew Melino, *America's Arctic Moment: Great Power Competition in the Arctic to 2050* (Center for Strategic and International Studies, March 2020); the various Atlantic Council reports on Arctic posture; and the Department of War's *Arctic Strategy* documents under both the Biden and Trump administrations.

[72] For the NATO Defence Planning Process, see NATO, "Defence Planning Process," available at https://www.nato.int/cps/en/natohq/topics_49202.htm.

across member states based on geography, capability, and strategic significance. A member state's Minimum Capability Requirements therefore reflect its position in the alliance rather than its absolute size.

Finland's accession to NATO in 2023 included specific assessment of its strategic significance for the alliance's northern flank, the Baltic approaches, and the High North. The accession process did not weight Finland's contribution by its population. The process assessed Finland's geographic position, its existing defense capabilities, its institutional capacity for total defense, and its strategic value for the alliance's collective posture. Finland's Minimum Capability Requirements as a NATO member reflect this strategic-significance-based assessment. The country's commitments include significant air policing contributions, naval presence in the Baltic and northern Atlantic, and substantial conventional defense capabilities that exceed what a population-weighted distribution would assign. Finland accepts these commitments because they correspond to Finland's strategic significance, and the alliance values Finland's contribution at the level of strategic significance rather than at the level of population.[73]

Estonia's case is sharper as Estonia's population of approximately 1.3 million represents less than 0.15 percent of the NATO total population. NATO's investment in Estonian defense is qualitatively larger than this population share would suggest. NATO Cooperative Cyber Defence Centre of Excellence is hosted in Tallinn. Enhanced Forward Presence battlegroups under NATO command are stationed in Estonia. Allied air policing missions operate from Estonian bases. NATO Multinational Division North headquarters operates with significant Estonian involvement. These investments reflect Estonia's strategic significance, its position on the Baltic, its border with Russia, its role as the eastern flank's most exposed member state, rather than its population. The alliance's framework treats strategic significance as the primary measurement criterion. The population-weighted analysis Estonia would receive under a Hazards-of-Place-style framework would underweight Estonia's strategic significance. NATO does not use the Hazards-of-Place-style framework for the alliance's defense planning, and the contrast is methodologically instructive.[74]

The Finnish Total Defence concept extends the strategic-significance logic to the internal structure of national defense planning. Finland's framework explicitly integrates municipal governance, private sector entities, civilian infrastructure operators, and citizen volunteer organizations into national defense exercises and standing planning processes. The integration is grounded in a specific institutional theory: small countries with significant strategic exposure require the entire society to be organized for defense because the country cannot afford the inefficiencies of compartmentalized defense planning. The Finnish Society for Security Studies and the Finnish Defence Forces conduct regular national defense exercises in which civilian sectors participate as primary actors rather than as auxiliary support. The Total Defence Course conducted by the Finnish National Defence College brings together senior officials from

[73] For Finnish Total Defence, see Government of Finland, "Security Strategy for Society" (most recent edition, 2017), and the Finnish National Defence College's Total Defence Course documentation. For analytical treatment, see Charly Salonius-Pasternak, "Finland's National Defence: An Overview," Finnish Institute of International Affairs Briefing Paper, various editions.

[74] For NATO investments in Estonia, see NATO, "Enhanced Forward Presence" and "NATO Cooperative Cyber Defence Centre of Excellence," available through NATO public documentation. For analytical treatment, see Andres Kasekamp, *A History of the Baltic States*, 2nd ed. (Palgrave Macmillan, 2018), and contemporary policy analysis from the International Centre for Defence and Security in Tallinn.

government, business, civil society, and the defense establishment in extended joint training that produces shared operational understanding across sectors that would not otherwise interact.

Estonia's broad societal defense framework extends similar logic to Estonia's specific case. The framework explicitly recognizes that defense in the contemporary threat environment requires civilian infrastructure operators, civil society organizations, and ordinary citizens to be integrated into defense planning at the operational level rather than at the rhetorical level. Estonia's response to the 2007 Russian cyberattacks produced institutional reforms that have been built into the broad societal defense framework over the subsequent two decades. The framework now integrates cyber defense, information operations resilience, critical infrastructure protection, and continuity of government planning into a unified national-defense architecture in which the Defence Forces are one actor among many.

The United States has not extended either the analytical framework or the institutional model to its own case. The Hazards-of-Place tradition's continuing influence in American emergency management reflects this. The NIPP's sector-specific structure reflects this. The strategic-capability tradition's aggregate national assessments reflect this. The combined effect is that the American analytical infrastructure produces conclusions about Alaska that the Finnish or Estonian analytical infrastructure would not produce about itself, and the operational consequence is that Alaska's strategic significance is dated in U.S. planning processes that determine investment, posture, and institutional architecture.

Measurement Framework for This Research

Six dimensions structure the assessment.

The first dimension is the share of national strategic capability physically located in the territory in question. For Alaska, this includes the share of U.S. ballistic missile defense capability; approximately ninety percent of the deployed Ground-Based Midcourse Defense interceptors are at Fort Greely. It includes the share of U.S. Pacific air power forward presence, JBER and Eielson together represent the principal Air Force presence between the West Coast and Japan. It includes the share of NORAD execution capability, the Alaska NORAD Region covers airspace from the Canadian border west across the Bering Strait. It includes the share of U.S. Coast Guard captain-of-the-port authority, Sectors Anchorage and Western Alaska together cover a coastal area larger than the entire continental U.S. East Coast. It includes the share of U.S. early-warning radar coverage of polar approaches, Clear Space Force Station is one of a small number of CONUS-relevant sites.

The second dimension is the degree to which the capability physically located in the territory depends on infrastructure that cannot be substituted at scale by infrastructure located elsewhere. For Alaska, this dimension assesses the substitutability of the Port of Alaska, Ted Stevens, the Cook Inlet energy infrastructure, the Anchorage communications nodes, and the institutional knowledge that operates these systems. The wargame's findings establish that none of these is substitutable at scale by infrastructure currently in place elsewhere.

The third dimension is the alliance and treaty obligations that depend on the territory's operational functioning. For Alaska, this includes NORAD obligations to Canada and through Canada to the broader Western alliance system, U.S. obligations under the Treaty of Washington as applied to North Atlantic

and Arctic security, U.S. commitments under the Indo-Pacific posture documents to Pacific allies, and U.S. participation in the Arctic Council and related multilateral arrangements. Alaska's operational functioning is a precondition for substantial U.S. obligations to multiple alliance and partner relationships.

The fourth dimension is the cascading effects across military, civilian, and economic systems if the territory's primary infrastructure degrades. Military effects include loss of forward air capability, degradation of missile defense readiness, disruption of NORAD execution, and inability to project power across the Pacific approaches. As one participant with USN experience noted, "[you] probably need a Carrier Strike Group (CSG) or two Expeditionary Strike Groups (ESGs) to cover the military gaps." Civilian effects include population mobility crisis, health system collapse, and humanitarian conditions that require either evacuation or external support at scales beyond current planning. Economic effects include disruption of trans-Pacific air cargo, disruption of strategic seaport function, and disruption of Trans-Alaska Pipeline operations. As such, cascading effects extend substantially beyond Alaska into national and global systems.

The fifth dimension is the adversary's strategic interest in the territory and the cost-imposition opportunities the territory's vulnerabilities provide. Russian and Chinese published Arctic doctrine treats the North American Arctic as a strategic competition zone. Russian information operations in Alaska Native communities are documented in the patterns the Red Team modeled in good faith. Chinese commercial and dual-use investments in Arctic research and infrastructure indicate sustained strategic interest. The wargame's gray-zone scenarios reflect documented adversary capability and intent. Adversary strategic interest in Alaska is substantial and that the cost-imposition opportunities are significant given Alaska's institutional vulnerabilities.

The sixth dimension is the time required to substitute for the territory's function if degradation occurs. For Alaska, the relevant substitution timeframes are measured in years for infrastructure investment and in decades for institutional knowledge replacement. Building deep-water port capacity at Nome or Port Clarence is a multi-year construction project even after authorization and funding. Establishing alternative trans-Pacific air cargo nodes capable of substituting for Ted Stevens would require multi-year facility development and air carrier route restructuring. Replicating the institutional knowledge held by the Port of Alaska's operating community, the Anchorage emergency management community, the Alaska National Guard, and the Coast Guard's Alaska sectors would require sustained recruitment and training programs across the decade timescale. Substitution time is long across all relevant categories, meaning that degradation of the existing substrate produces consequences that cannot be quickly mitigated.

Alaska's strategic significance is qualitatively and quantitatively decoupled from its population. The assessment is not a single number on a national distribution. It is a structured argument grounded in specific evidence about specific dependencies. The measurement approach used herein is essentially the same approach NATO uses for Finland and Estonia, the same approach Finnish Total Defence uses internally, and the same approach Estonian broad societal defense uses.[75]

[75] Excepting reference to considering adaptive capabilities for problem solving to be a novel way of measuring state power derived from a separate doctoral contribution to the international relations literature on state power (Iverson, 2026 anticipated).

Cost-imposition and the institutional architecture as adversary target.

The Red Team's strategic logic during the hot wash captured the underlying dynamic with unusual clarity. Cost-imposition does not require any particular adversary action; it requires that the defender's institutional response architecture produces disproportionate costs in response to ambiguous threats. The Red Team's only requirement was to generate ambiguity. The defender's institutional architecture did the rest.

This formulation reframes how the strategic competition the wargame modeled should be understood. The standard reading of gray-zone activity treats it as a problem of attribution and response, adversaries conduct ambiguous activities that fall below the threshold of conventional response, and the defender's challenge is to identify the activity correctly and respond proportionately. The wargame's findings suggest a different reading. The adversary's target is not the specific operational asset the ambiguous action ostensibly threatens. The adversary's target is the defender's institutional architecture itself. As a Red Team participant summarized, “Our real motivation was to discredit the US government and use the environmental and institutional judo to do the rest of the work.” The backpack near the Port of Alaska's valve operating system was not an attempt to damage the valve. It was an attempt to demonstrate that the defender's response architecture would lock in disproportionate costs the moment ambiguity was generated. The cost-imposition is what the adversary obtains; the specific operational consequence is secondary.

This reading has direct implications for how the United States should approach Arctic gray-zone activity. The defender's institutional architecture is the variable the adversary is actually targeting. The architecture's design, the FBI prioritization dynamic, the FPCON elevation procedures, the MARSEC level mechanics, the counter-UAS authority structure, the cleared-uncleared information asymmetry, determines how much the adversary can extract from any given ambiguous action.[76] Reforms to that architecture are therefore the most consequential defensive investments. The recommendation that information operations, command architecture, and logistics must stand up simultaneously on Day 1 of any crisis is the operational expression of this claim. Sequential standup creates exactly the windows adversaries exploit. The institutional habit of solving one problem before moving to the next is the institutional vulnerability adversaries target. Parallel standup denies the adversary the gap between when an ambiguous action occurs and when the defender's full architecture is in place. The wargame demonstrated the principle through gameplay; Alaska National Guard representative articulated it as the most consequential learning from the decision-making process.

The Finland and Estonia comparison.

The wargame's finding that the U.S. security and municipal governance architectures rarely interact, emerged through the co-lead’s framing during the post-game meeting, has a specific institutional alternative in the Finnish total defense doctrine and the Estonian broad societal defense framework. Both countries have systematically integrated municipal governance with national security in ways the U.S. system has not. Finnish municipalities have explicit emergency preparedness obligations coordinated with

[76] Day 2 afternoon transcript, counter-UAS authority discussion. The five categories of authorized actors are codified in 14 CFR 124/130I and related authorities. The recently-added tribal authority requires FBI-coordinated training before authorized engagement.

the Finnish Defence Forces. Estonian municipalities and private sector entities participate in regular national-defense exercises that include civilian infrastructure resilience as a primary mission set.[77] Both countries treat civilian-military integration as a peacetime institutional investment rather than as a crisis-time improvisation.

The comparison illuminates what the United States has not done. The wargame's repeated finding that the room contained people who would not normally coordinate, working through coordination problems for the first time under simulated stress, reflects the absence of standing institutional structures for civil-military coordination at the operational level in the United States. The Department of War has its planning processes. FEMA has its planning processes. State emergency management agencies have theirs. Private sector critical infrastructure operators have theirs. Tribal governments have theirs. None of these planning processes routinely intersect during peacetime in ways that produce shared operational understanding, established trust, and rehearsed coordination. The United States expects crisis-time improvisation to produce what Finland and Estonia produce through peacetime institutional design. The recommendation is to develop institutional mechanisms that produce comparable effects to Finnish total defense and Estonian broad societal defense, adapted to the American federal system. Alaska is the proving ground for this development because Alaska is where the institutional gap is most operationally consequential.

VIII. RECOMMENDATIONS

The wargame revealed a set of requirements that any adequate Arctic homeland-defense posture would have to meet, along with a demonstration of where the current posture falls short.

A. Logistics redundancy for an irreplaceable hub

The wargame showed that Alaska's dependence on a single deep-water port with gantry and roll-on/roll-off capacity and on a single primary air-cargo hub, left both humanitarian and homeland-defense logistics with no fallback once those nodes were degraded. The Naval Station Everett deficit confirmed that maritime substitution from outside the state is not available on operationally relevant timelines: Everett holds guided-missile destroyers rather than amphibious lift, and the nearest amphibious capability is roughly two weeks of cold-weather transit. This is the operational condition the wargame demonstrated should be assessed at the level of U.S. strategic significance rather than at the level of regional population.

Force-posture and infrastructure assessments should weigh how to introduce redundancy into Alaska's deep-water port and heavy-air-cargo capacity. The options include additional hardened port infrastructure,

[77] For Estonia's broad societal defense framework, see Estonian Ministry of Defence, "National Defence Strategy of Estonia" (most recent edition), and the Estonian Defence League's documentation of citizen integration into defense planning. For analytical treatment of the 2007 cyberattack response and its institutional consequences, see Tarmo Soomere, ed., *Cyber War in Perspective: Russian Aggression against Ukraine* (NATO Cooperative Cyber Defence Centre of Excellence, 2015), and subsequent CCDCOE publications.

of which Nome (already underway) and Port Clarence are the most studied candidates.[78] However, it is important to note the relevant need for access to highway/rail and air connections; additional airfield capacity able to handle heavy-lift aircraft; and surface-transportation alternatives that do not depend on the existing single-road, single-port, single-hub configuration. The wargame establishes the requirement for redundancy but the specific mix of port, air, and surface investment is a question for the assessment process, informed by the existing Congressional Research Service and Army Corps of Engineers work on deep-draft feasibility.

B. Workforce resilience

The personnel finding established that Alaska's response architecture depends on a workforce broader than the conventional responder categories. That workforce is smaller in absolute numbers than its Lower 48 equivalents, and current planning wrongly treats it as replaceable from outside the state through surge mechanisms.

Federal and state mechanisms currently extend protected employment status, deployment authority, and family support to defined responder categories: sworn law enforcement, certified emergency medical services, certified firefighters, designated emergency management personnel, and military responders. The wargame established that the response also depends on workers outside those designations. The longshoremen and stevedores at the Port of Alaska, the deicing crews at Ted Stevens and Joint Base Elmendorf-Richardson, the snowplow and road-maintenance crews, the fuel-truck drivers on the Valdez and North Pole routes, the turbine operators at Beluga, the water and wastewater operators, the communications technicians at the satellite ground stations and cable landings, the air traffic controllers, the harbor pilots, the Alaska Marine Highway crews, the bypass-mail pilots, the refinery workers, and the hospital support staff are all operationally essential and all invisible to the existing frameworks. Assessments should weigh whether and how to extend protected status to these categories during defined Arctic crisis conditions, whether through Alaska-specific designation under existing federal authorities, through state legislation mirroring federal responder protections, or through both.

The existing support for conventional responders includes pre-positioned childcare, family communication systems, and post-deployment assistance, with the Department of War's installation-level infrastructure the most developed example and civilian state and municipal support the least.[79] Assessments should weigh how to scale that support to the direct-labor workforce that operates the physical infrastructure on which response depends, including childcare scaled to that population, communication that survives the degradation the wargame underscored as plausible, and the employer-to-provider arrangements that let support activate quickly. Those arrangements would require coordination

[78] For the Port of Nome modernization and Port Clarence assessments, see U.S. Army Corps of Engineers Alaska District, "Arctic Deep Draft Port" feasibility studies and current project documentation. The current Port of Nome modernization is funded under the Bipartisan Infrastructure Law and the Disaster Relief Supplemental Appropriations Act.

[79] For documentation of Alaska's military installations and their roles, see U.S. Indo-Pacific Command, "Alaskan Command," at https://www.alcom.mil; U.S. Northern Command, "Alaska NORAD Region," at https://www.norad.mil; Missile Defense Agency, "Ground-Based Midcourse Defense (GMD)," at https://www.mda.mil. The 11th Airborne Division was reactivated in June 2022 as a unit-of-action realignment specifically structured for Arctic operations.

among federal agencies, the state, municipal governments, and the private and quasi-public employers who hold the employment relationships, which the current structure does not provide for.

Alaska's small population produces a pattern in which the same small group staffs multiple critical functions, a strength under most conditions and a vulnerability under the conditions the wargame demonstrated. Assessments should weigh sustained investment in cross-training that builds depth for each critical function rather than relying on a handful of irreplaceable individuals. Federal funding for apprenticeship and certification in the identified direct-labor categories, state workforce-development investment aimed at the consolidation problem, and coordination with major employers to support cross-training during normal operations are the available levers; the University of Alaska system, the Alaska Vocational Technical Center, and the relevant labor organizations are the natural partners. The horizon is multi-year, because the institutional knowledge the wargame identified as critical takes years to develop in any worker and decades across a workforce.

The architecture must be able to function from its own standing resources for periods longer than current planning assumes, and in some scenarios indefinitely. Assessments should weigh revising standing emergency plans so the no-surge condition becomes the baseline rather than an unmodeled background assumption. That revision would reach pre-positioning requirements, which scale up sharply when outside stocks cannot be counted on; standing personnel arrangements, which need Alaska-resident depth where current planning relies on Lower 48 surge; and the coordination functions current planning expects Lower 48 personnel to support.

C. Distributed energy and hardened communications

The wargame found, through the play, that Anchorage's electricity depends on natural-gas turbines vulnerable to the same volcanic ash that grounds aviation. It also exposed an uneven communications picture: redundant in Southcentral, thin and satellite-dependent in rural Alaska, with civilian satellite throughput subject to government prioritization in a crisis. These single-mode dependencies failed, or threatened to fail, under the modeled conditions.

Resilience assessments should weigh how to reduce them. On energy, the options include distributed generation that does not share the turbines' ash sensitivity, hardened backup capacity, and fuel-supply arrangements that account for the winter weight restrictions the wargame identified as the binding constraint. On communications, the options include hardened terrestrial redundancy, pre-arranged prioritization for emergency users on satellite systems, and rural communications that do not collapse to a single degraded path. The requirement is reduced single-point dependency. The engineering is for the assessment to determine.[80]

D. Tribal and Alaska Native engagement as pre-crisis infrastructure

The wargame tested, and complicated, the assumption that Alaska Native communities would uniformly refuse adversary humanitarian overtures. Structured play found the assumption of uniform refusal to be

[80] For Arctic renewable energy research and deployment, see University of Alaska Fairbanks, Alaska Center for Energy and Power, "Arctic Energy Office" publications, available at https://acep.uaf.edu. For Alaska Industrial Development and Export Authority programmatic infrastructure, see https://www.aidea.org.

wrong: specific communities with specific historical and cultural ties would respond differently from one another, and federal refusal at the State Department level does not prevent acceptance at the community level. Play also showed that standard disaster planning ignores subsistence cycles, so that a failed autumn hunt becomes a winter food-security problem the planning baseline never captures.

Relationships with Alaska Native organizations function as response infrastructure, and infrastructure has to be built before a crisis rather than improvised during one. Assessments should weigh how to invest in those relationships during peacetime: sustained engagement with tribal organizations, planning that incorporates subsistence cycles, and the kind of formal pre-crisis coordination that lets crisis-time response rest on existing trust. The wargame establishes that uniform-cooperation assumptions are a planning failure. The form the engagement takes is for the relevant authorities and communities to determine together.

E. Pre-crisis legal and authority architecture

The wargame repeatedly showed that the authorities governing domestic crisis response do not bend under operational pressure: FAA temporary flight restrictions and their conversion to defense TFRs, Posse Comitatus, Stafford Act thresholds, tribal sovereignty, and the layered counter-UAS authority structure with its five categories of authorized actors.[81] These are legal architectures. Commanders cannot work around them at the moment.

The requirement, in one military participant's words, is that they be worked at the policy level before a crisis, so commanders hold pre-delegated authorities rather than improvising workarounds when speed determines outcomes. Assessments and interagency processes should weigh how to clarify and pre-delegate the relevant authorities ahead of time. The counter-UAS authority structure the wargame examined is the clearest example of where this is urgent. The requirement is pre-crisis legal clarity; the specific delegations, and the bodies that issue them, are for NORTHCOM, the Coast Guard, FEMA, and the civilian authorities to work out.

F. Planning for parallel rather than sequential execution

One after-action submission gave one of the clearest findings: the response identified the right actions early, dual-status command activation, communications, aviation assumptions, Alaska Marine Highway retasking, and executed them late, with each delay compounding. Standard incident-command architecture is built for sequential prioritization, and in a strategic-competition environment that sequencing opens windows an adversary can exploit. The finding is that information operations, command architecture, and logistics have to stand up at once, with no phase in which any can be safely deferred.

Doctrine and planning assessments should weigh how to build parallel execution into Arctic crisis planning rather than defaulting to sequence. The options include revised incident-command guidance for

[81] Day 2 afternoon transcript, counter-UAS authority discussion. The five categories include Department of War personnel, Department of Homeland Security personnel, specifically designated law enforcement, certain critical infrastructure protection personnel, and recently added tribal authorities under 14 CFR 124/130I requiring FBI-coordinated training

compound scenarios that pair strategic competition with disaster, pre-authorized parallel activation triggers, and exercise design that rehearses simultaneous stand-up. The institutional default should shift toward parallelism where the operating environment demands it.

G. Formalized private-sector and voluntary-organization arrangements

The wargame showed essential functions migrating to unlikely homes because the formal structures were not built for the actual crisis.[82] A fusion center oriented almost entirely to narcotics was eyed for ad hoc transformation into an intelligence-fusion role; mass care and feeding depended on voluntary organizations whose contribution was real but not formally integrated. Government resources alone did not meet the modeled survivor needs, which matches the state's own Emergency Operations Plan assumption that volunteer, non-governmental, and faith-based organizations will carry a significant share of mass care.[83]

Assessments should weigh how to formalize these arrangements before a crisis rather than building them during one. That includes standing agreements with the private-sector operators of critical infrastructure and with the voluntary organizations active in Alaska disaster response. The dependencies the wargame exposed are real; what they lack is formalization in advance.

H. A follow-on wargame

Several of the most consequential findings are hypotheses the wargame produced rather than facts it established: the natural-gas-turbine vulnerability, the differential community response, the no-scalable-alternatives observation. The methodology section is candid that gaps in the roster, above all the absence of tribal and Alaska Native participation, limit what this iteration can confirm. A follow-on iteration is therefore itself a recommendation. A summer-variant scenario would test seasonality against the January baseline; the participation this round lacked, from tribal organizations, FEMA, state emergency management, and major utilities, would let the differential-response and no-alternatives findings be tested directly; and the gray-zone dimension would be present from the opening move rather than introduced midway.[84] The point is to validate these hypotheses through better-resourced play before they harden into planning assumptions.

IX. CONCLUSION

The wargame tested whether the State of Alaska could sustain homeland defense and mass-care operations simultaneously under conditions of severe environmental degradation and adversary gray-zone

[82] The literature on prosocial behavior under crisis conditions provides relevant analytical grounding for the volunteerism dynamic the wargame did not adequately test. See Jamil Zaki, "Catastrophe Compassion: Understanding and Extending Prosociality Under Crisis," Trends in Cognitive Sciences 24, no. 8 (2020): 587-589. Zaki's finding that crisis tends to surge prosocial behavior rather than collapse it has direct operational implications for response architecture design; the question is whether institutional architecture can absorb and direct the surge effectively.
[83] Ibid.
[84] USCG representative, after-action submission. Quoted with permission

activity. The answer the wargame produced is that current institutional architecture and physical infrastructure are not adequate to the task. The state's primary logistics nodes are irreplaceable in their specific functions. The personnel who operate those nodes are subject to absenteeism rates that current planning consistently underestimates, with the personnel category itself extending well beyond conventional first responders to include the direct labor workforce on which all response operations depend. The evacuation architecture is configured for short-duration events rather than sustained Arctic crises and runs into physical infrastructure carrying capacity limits that traffic management practice cannot eliminate. The information environment includes vulnerabilities that adversaries exploit at low cost. The institutional tempo defaults to sequential processing that adversaries exploit through parallel operations. The Lower 48 surge support, private sector commercial response, and voluntary organization mobilization that current planning treats as reliable face constraints from concurrent demands that have intensified in recent years.

Alaska's strategic significance is decoupled from its population in ways that current American analytical frameworks cannot represent. The Hazards-of-Place tradition, the FEMA National Risk Index, the National Infrastructure Protection Plan's sector-specific implementations, and the strategic-capability tradition's aggregate national assessments all systematically miscategorize the Alaska case. The miscategorization is not an error within any individual framework. It is the cumulative effect of analysis developed against a problem space that does not match the problem space of Arctic homeland defense. The frameworks were developed for environments in which population size approximated strategic stakes, environmental events were episodic, and strategic competition occurred elsewhere. None of these conditions holds for the Arctic. The infrastructure American emergency management uses is overdue for refresh.

The alternative approach the report develops is not novel in its own right. NATO's planning frameworks for Finland and Estonia, the Finnish Total Defence concept, and the Estonian broad societal defense framework all use strategic-significance-weighted assessment grounded in geography, threat exposure, and alliance position rather than in population at risk.[85] These frameworks treat civilian infrastructure operators, civil society organizations, and ordinary citizens as integral to national defense rather than as auxiliary support. They build the relationships and institutional capacity that crisis response requires during peacetime rather than expecting crisis-time improvisation to produce them. The American institutional tradition has not extended this approach to its own Arctic territory. The wargame demonstrates that the absence is operationally consequential.

Alaskan airport and deep-water cargo port infrastructure is limited and brittle, pointing to the need for investing in expanded and redundant port and airport infrastructure. Distributed energy generation and hardened communications address the substrate vulnerabilities the gameplay underscored. Workforce protection, family support, and resilience as operational requirements extended across the direct labor categories the response architecture depends on addressing the personnel finding in its full scope. Direct tribal engagement during peacetime addresses the cultural and community-level dynamics the wargame could not adequately test without direct tribal participation. Pre-crisis legal architecture addresses the

[85] For Finnish Total Defence, see Government of Finland, "Security Strategy for Society" (most recent edition, 2017), and the Finnish National Defence College's Total Defence Course documentation. For analytical treatment, see Charly Salonius-Pasternak, "Finland's National Defence: An Overview," Finnish Institute of International Affairs Briefing Paper, various editions

authority constraints that do not bend under operational pressure. Institutional reform toward parallel rather than sequential planning addresses the tempo asymmetry adversaries currently exploit. Formal private sector and voluntary organization arrangements address the parallel limit on commercial and philanthropic surge that current planning has not adequately recognized.

The investments recommended are not ones that can be substituted by surge mechanisms from outside Alaska. The existing federal architecture is increasingly constrained by its own demands and cannot reliably substitute for Alaska-located capacity. The private sector commercial response cannot be relied upon to produce volume that contradicts commercial cost-benefit calculations under sustained Arctic crisis conditions. The voluntary organization mobilization cannot be relied upon to produce capacity that concurrent national demands are simultaneously consuming. Pre-positioned investment in Alaska is operationally necessary because the alternatives, such as a Lower 48 governmental surge, private sector commercial response, voluntary organization mobilization, cannot be relied upon. Alaska's strategic significance is decoupled from its population by a substantial margin, and the federal, commercial, and voluntary surge architecture that might in principle compensate for under-investment in Alaska-located capacity is itself increasingly constrained. Alaska's standing capacity must be developed to a substantially higher standard than current planning treats as adequate.

Alaska is the proving ground for whether the United States can recognize that strategic significance can exceed population weight by orders of magnitude. The state's 740,000 residents are the people whose presence makes the strategic substrate functional. The substrate hosts approximately ninety percent of the GBMD interceptors that defend the entire United States from intercontinental ballistic missile threats. It hosts the principal Pacific air power forward presence between the West Coast and Japan. It executes NORAD's responsibility for the western and northern approaches to North America. It exercises Coast Guard captain-of-the-port authority across a coastal area larger than the entire continental U.S. East Coast. It handles over half of all air cargo flown from Asia to North America, ranking fourth among global cargo airports by tonnage. It holds Strategic Seaport designation for the only mainland Alaska port capable of supporting military deployment and sustainment.[86] None of these strategic functions is substitutable at scale by infrastructure located elsewhere. None of them has the visibility in current frameworks that the wargame's findings demonstrate as warranted.

Alaska is also the proving ground for how the United States operates in degraded environments more broadly. The strategic competition pressures documented throughout will continue. The institutional vulnerabilities the wargame exposed will continue until they are addressed. The Arctic is where the cumulative pressure is most operationally consequential because the geography is most unforgiving, but the cumulative pressure is not unique to the Arctic. The methodological argument, that American analytical infrastructure for assessing homeland-defense-relevant vulnerability needs refresh to accommodate the conditions that have emerged over the past two decades, applies beyond the specific Alaska case. The adaptive problem-solving framework the author develops in her separate doctoral

86 Anchorage Economic Development Corporation. (2023). Ted Stevens Anchorage International Airport economic impact study. https://aedcweb.com/wp-content/uploads/2023/04/Ted-Stevens-Anchorage-Economic-Impact-Study-2023.pdf

research is one contribution to this wider project; the wargame is one empirical contribution to building the assessment criteria American homeland defense increasingly requires.[87]

The Alaska wargame highlighted the Arctic's unique operational environment, yet the single-point-of-failure dynamics observed in Anchorage mirror a national pattern of vulnerable hub-and-spoke infrastructure. The 2025 National Security Strategy's reframing of strategic priorities toward the Western Hemisphere provides a rhetorical basis for the report's recommendations. The coherence of this Western Hemisphere strategic concept hinges on Alaska's operational functioning. As the northwestern anchor of the Western Hemisphere, the United States cannot achieve strategic primacy in the region if its Alaskan Arctic territory is operationally compromised. The 2025 NSS framework places Alaska's operational functioning at the center of stated administration priorities, even though the document does not name it directly.

[87] Iverson (forthcoming 2026). The framework's central arguments are that adaptive problem-solving capacity is the forgotten core of state power and that capability-based assessments underweight this variable and are developed at length in that work. The framework's specific articulation, its scholarly grounding in the international relations and American politics literatures, and its application to comparative state-capacity cases are addressed in the dissertation. This research applies the framework to the case of Arctic homeland defense, using the wargame's findings to demonstrate how the framework illuminates a vulnerability the dominant analytical traditions miss.

Appendices

APPENDIX A | Wargame Participating Agencies and Organizations

The wargame was conducted under the Chatham House Rule. Individual participants are not named in this appendix or in the body of the report. Participating agencies and organizations are listed below by team assignment and role. Three organizations whose representatives were invited but unable to attend the executed wargame are noted at the end of this appendix.

Blue Team (Operational Stakeholders)

Port of Alaska — Operations and Security Support — Port Safety Coordinator

Municipality of Anchorage — Unified Command, Emergency Operations Center Director — Emergency Programs Manager

Launch Alaska — Deputy Director, Transportation — Chief Deployment Officer

Ted Stevens Anchorage International Airport — Airport Safety Specialist

Alaska National Guard — Joint Staff

DLA Energy Americas North — Quality Assurance Representative

U.S. Coast Guard — Sector Western Alaska and U.S. Arctic — U.S. Coast Guard Fellow, Ted Stevens Center for Arctic Security Studies — Ted Stevens Center Fellow, U.S. Coast Guard

U.S. Army — U.S. Army Military Fellow, Ted Stevens Center for Arctic Security Studies

DHS Cybersecurity and Infrastructure Security Agency — Protective Security Advisor

Red Team (Adversary Doctrine Modeling)

Three subject-matter experts modeling Russian and Chinese gray-zone doctrine, drawn from academic and federal logistics backgrounds. Red Team contributions are reported under collective non-attribution per the Red Team participation agreement.

White Cell (Adjudication and Scenario Management)

Idaho National Laboratory — Senior Critical Infrastructure Analyst

Homeland Defense Institute — Wargame lead author

Wargame Co-Lead — Participating in personal capacity

Not Represented in Executed Wargame

The following organizations were invited to participate but their representatives were unable to attend the executed wargame in April 2026. Their absence is noted because their inclusion would have addressed specific gaps in the wargame's coverage.

University of Alaska Fairbanks International Arctic Research Center | Alaska Army National Guard G3 Operations Staff | Idaho National Laboratory (second analyst position)

Significant Gaps in Participant Composition

There were broader gaps in the participant composition that the wargame did not address. These include the absence of direct Alaska Native and tribal representation; the absence of FEMA representation; the absence of state-level emergency management beyond the white cell's stand-in for the State Emergency Operations Center; the absence of the American Red Cross and other major nongovernmental humanitarian organizations; the absence of major private utilities (Chugach Electric, ENSTAR Natural Gas, Hilcorp); the absence of major Alaska Native regional and village corporations; and limited representation from the air and maritime cargo industries.

APPENDIX B | Wargame Session Summaries

This appendix provides session-by-session summaries of the wargame as executed in Anchorage, Alaska, in April 2026. The summaries serve as a navigable record for readers who want to trace specific findings back to the gameplay sessions that produced them. Detailed transcript excerpts illustrating specific findings appear in Appendix C.

Day 1, Morning Session | Introduction and Move 1 Setup

The morning session opened with introductions by the wargame leads and a briefing on the scenario premise: a notional Shishaldin volcanic eruption (Volcanic Explosivity Index 4, modeled on the 2010 Eyjafjallajökull event) producing a persistent ash plume that had been disrupting Alaska operations for approximately two weeks at the moment Move 1 began.

The white cell delivered the situational brief covering the volcanic ash trajectory across the Aleutians, Cook Inlet, and extending north toward Fairbanks; the prevailing wind patterns; the current operational status of major aviation, maritime, and ground transportation infrastructure; and the standing humanitarian assistance and disaster response baseline. Participants were grouped at tables according to a Blue Team organizational structure that mirrored the Anchorage emergency operations architecture they would in fact be expected to staff during a real event.

The session ended with the room oriented to the scenario and ready to begin Move 1 gameplay. Several participants noted during the transition that the upfront ground-truthing felt thin for a two-week-into-crisis scenario and that they would have benefited from more detailed scenario specification on baseline conditions.

Day 1, Afternoon Session — Move 1 Execution

Move 1 focused on the baseline humanitarian assistance and disaster response operational picture at approximately day fourteen of the volcanic ash event. The white cell injected a multi-vessel collision incident near the Barren Islands involving a Chinese-flagged commercial vessel, which provided the maritime emergency response component to the move. Participants worked through coordination of the marine casualty response while simultaneously managing the ongoing population and infrastructure stress produced by the sustained ash conditions.

The afternoon session surfaced the wargame's central operational finding through participant articulation rather than designed inject. The DHS CISA Protective Security Advisor framed Alaska's infrastructure vulnerability as a "spider web of single points of failure" with the Port of Alaska at its center.[88] Participants from the Port of Alaska, Ted Stevens Airport, and the Municipality of Anchorage Emergency Operations Center confirmed the framing from their operational vantage points. The natural gas turbine vulnerability emerged through gameplay when the Launch Alaska Deputy Director of Transportation surfaced the combined-cycle versus simple-cycle distinction at the Beluga generation site and the subsequent cascading dependency through electricity, water, fuel pumping, and habitable temperature maintenance. The white cell adjudicated the finding through dice and through subsequent participant outreach to industry contacts, confirming the operational implications and incorporating the vulnerability into subsequent moves as a sustained condition.

The session also produced substantial discussion of personnel availability, with the DLA Energy Quality Assurance Representative articulating the family-first framing drawn from the federal logistics community's COVID-19 period experience and the Alaska National Guard representative noting the Mount Spurr family evacuation precedent. The Municipality of Anchorage Unified Command EOC Director articulated the operational implication for response architecture: planning must accommodate family-first behavior rather than ignore it.

The session ended with the room having established the operational baseline, the central vulnerability framing, and the personnel and family-care planning implications. The hot wash brief at the close of Day 1 captured the upfront-ground-truthing limitation that several participants identified.

Day 2 | Morning Session | Move 2 Execution

Move 2 elevated the operational stress through multiple injects designed to test institutional response under compound conditions. The white cell injected the Canadian Navy HMCS Vancouver deployment from CFB Esquimalt, the Chinese Coast Guard vessel Nansha approach with humanitarian framing toward the Aleutians, and the Russian icebreaker repositioning toward Nome and Kotzebue invoking Chukotka cultural ties and Russian Orthodox religious heritage. The Force Protection Condition was elevated to CHARLIE during the session.

The HMCS Vancouver inject illustrated the structural finding about NORAD's maritime mission gap. The Alaska National Guard representative and the USCG Sector Western Alaska representative both

[88] Statement by participant from DHS CISA, Move 1, Day 1, April 2026. Quoted with permission.

identified immediately that the Canadian deployment was unilateral rather than through NORAD because the alliance command does not have a standing maritime mission.[89]

The Chinese Coast Guard Nansha approach was adjudicated through dice for marine casualty in the ash plume. The roll came up 8 of 12 (1-8 indicating breakdown), producing the casualty. The subsequent discussion of U.S. rescue response and its strategic implications produced the finding on how adversary-degraded operating performance creates U.S. operational opportunities when the U.S. rescue and recovery capability outclasses the adversary's.

The Russian icebreaker scenario was the most substantial element of Move 2. The white cell adjudicated through structured dice rolls: subsistence hunt success in the affected communities (9 of 12, indicating adequate hunts), whether any community members boarded the Russian vessels (3 of 12, with threshold 1-2 for yes; no boarding occurred), and the legal status of aid offered without boarding (effectively certain acceptance because federal authority does not bind community-level decisions beyond 12 nautical miles). The exchange produced the report's findings on differential tribal community response to adversary humanitarian aid and on the limits of federal authority over community-level aid acceptance.

The backpack incident at the Port of Alaska's valve operating system was injected during the morning session and produced the cost-imposition finding. The Anchorage Police Department participant, the Port of Alaska Operations and Security Support participant, and the USCG representative worked through the institutional response sequence: FBI engagement, MARSEC level elevation, FPCON adjustment, and personnel diversion. The white cell subsequently determined the backpacks did not contain explosives. The security posture elevations the initial response triggered did not unwind. The cost was locked in at the moment of initial response, driving the textbook cost-imposition pattern.

Day 2 | Afternoon Session | Move 3 Execution and Hot Wash

Move 3 introduced gray-zone probing activity with suspected sabotage at the remaining operational fuel terminal at the Port of Alaska. The Defense Support of Civilian Authorities (DSCA) framework was activated. FPCON was elevated to Delta. Participants worked through coordination of the elevated security posture while simultaneously managing the sustained humanitarian assistance and disaster response operational picture under conditions of compounding institutional stress.

The afternoon session revealed the institutional tempo finding most clearly. Participants identified across multiple decision points that the response architecture was working through issues sequentially when parallel processing was required. The DSCA activation, the communications issues, the aviation optimism assumptions, and the Alaska Marine Highway retasking were each identified relatively early but executed substantially later. The pattern was visible to participants in real time but did not produce real-time correction.

The hot wash at the close of Day 2 produced the Red Team's strategic logic articulation. The Red Team explained the decision not to deploy cyber capabilities (environmental degradation had produced the

[89] Day 2 morning transcript, Canadian Navy injection and subsequent discussion. The military participants' immediate recognition that NORAD lacks a standing maritime mission is captured in the transcript and is consistent with published NORAD doctrine.

operational effects cyber would have produced), the strategic logic of the backpack incident (cost-imposition obtained from the defender's institutional response regardless of actual threat content), the Russian Orthodox angle in Alaska Native community targeting (real cultural ties exploitable through information operations on the documented Russian playbook), and the targeting of the defender's institutional architecture rather than specific operational assets.

The hot wash also captured the design lessons that the after-action submissions subsequently developed in detail. The upfront-ground-truthing limitation, the weather-data-fidelity gap, the late introduction of the gray-zone dimension, and the value of the three-move structure with end-of-move reflection products were all developed during the hot wash and developed in the after-action material.

Post-Game Synthesis Meeting | April 25, 2026

A post-game synthesis meeting between the wargame lead author and the wargame co-lead, produced the framing that organizes the report. The meeting focused on the relationship between the wargame's findings and the broader strategic questions about Arctic homeland defense, the Finland and Estonia institutional comparison, the critique of the dominant emergency management frameworks, the operational practice versus reality tension in evacuation planning, and the application of the adaptive problem-solving framework from the lead author's separate doctoral research to the Alaska case.

APPENDIX C | Selected Transcript Excerpts

This appendix presents curated excerpts from the wargame transcripts that document the basis for the report's findings. Excerpts are organized by finding category and are attributed by role and organization per the Chatham House Rule applied throughout the project. Each excerpt is presented with brief context identifying the session, the gameplay moment, and the report finding the excerpt supports. The full transcripts are maintained as internal records of the Homeland Defense Institute and are available to authorized readers who wish to verify the curated excerpts.

C.1 The "No Scalable Alternatives" Finding

The report's central claim that Alaska has no scalable alternatives if its primary infrastructure fails was articulated most concisely by the DHS CISA Protective Security Advisor during Move 1 on Day 1.

Day 1 afternoon, during Move 1 discussion of port and airport vulnerabilities.

DHS CISA Protective Security Advisor: "Alaska is a spider web of single points of failure. The very center of that web is the Port of Alaska. If you knock the Port out, you've knocked out the state. If you knock Ted Stevens out, you've knocked out the state. If you knock both out, which is essentially what this scenario does, you don't have a state anymore in any functional sense."

[Subsequent discussion among Port of Alaska Operations and Security Support, Ted Stevens Airport Safety Specialist, and Municipality of Anchorage Unified Command EOC Director confirmed the substantive accuracy of this framing from operational vantage. The Port participant noted that the gantry crane configuration cannot be substituted by alternative Alaska ports without infrastructure that does not

currently exist. The Ted Stevens participant noted that no other Alaska airport approaches Ted Stevens cargo throughput capacity.[90] The Anchorage EOC Director noted that the cascading effects on mass care, medical, and population mobility flow directly from the Anchorage hub's degradation.]

C.2 The Natural Gas Turbine Vulnerability

The natural gas turbine vulnerability emerged through gameplay rather than through a designed scenario, raised by the Launch Alaska Deputy Director of Transportation during Move 1 adjudication.

Day 1 afternoon, Move 1 adjudication of operational consequences of ashfall on Cook Inlet energy infrastructure.

Launch Alaska Deputy Director of Transportation: "The combined-cycle turbines at Beluga are more sensitive to particulate ingestion than the simple-cycle units. Under sustained ashfall, the combined-cycle generation goes offline. The simple-cycle units at Beluga can continue with intermittent filter clearing, and Bradley Lake hydroelectric continues, and Eklutna continues, but the combined capacity drops below what the Anchorage Bowl needs. You're looking at sustained brownout conditions across the Anchorage area."

[After the participant's outreach to industry contacts during the white cell's adjudication process, the finding was confirmed and the cascading implications were articulated.]

Launch Alaska Deputy Director of Transportation: "Without natural gas supply, there is no electricity in Anchorage. Without electricity, the wastewater pump stations stop. The fuel pump stations stop. The water treatment infrastructure progressively degrades. The buildings cannot maintain habitable temperatures. This is not a power outage. This is a habitability cascade."

[The white cell adjudicated the finding through dice and through participant outreach to industry contacts. The vulnerability was incorporated into Move 2 and Move 3 gameplay as a sustained condition rather than as a one-time inject.]

C.3 Personnel Absenteeism and the Family-Care Driver

The personnel absenteeism finding is grounded in participant statements that compounded across multiple sessions. The DLA Energy Quality Assurance Representative's framing, drawing on the federal logistics community's COVID-19 period experience, established the institutional weight of the family-care driver. The Alaska National Guard representative's Mount Spurr family evacuation reference established the pattern. The Municipality of Anchorage Unified Command EOC Director articulated the operational implication for the response architecture.

Day 1 afternoon, discussion of essential personnel availability under sustained crisis conditions.

[90] For Ted Stevens Anchorage International Airport's cargo statistics and trans-Pacific routing role, see Ted Stevens Anchorage International Airport, "About ANC," at https://dot.alaska.gov/anc. The airport consistently ranks among the top five global airports by cargo tonnage.

DLA Energy Quality Assurance Representative: "In disasters, family comes first. This is not theoretical. We saw it during COVID. We had federal logistics positions that we could not fill because the people in those positions had family obligations they had to attend to first. The federal logistics infrastructure, which is supposed to be resilient to exactly these kinds of conditions, faces sustained workforce competition during crisis. If we are honest about planning for a sustained Arctic crisis, we have to plan for the family-first reality, not against it."

Alaska National Guard representative: "When Mount Spurr erupted, my own family was among the first to evacuate the Glenn Highway. I was on the Guard call list. My family was on the road. This is what happens. The Emergency Operations Plan acknowledges this. Thirty-five to forty percent absenteeism among essential personnel is the planning baseline. It is not a worst case. It is the case."

Municipality of Anchorage Unified Command EOC Director: "The question for us is not whether to plan for family-first behavior. The question is whether the response architecture accommodates it or tries to ignore it. If we plan for family-first, we set up the institutional infrastructure that makes the response work despite the absenteeism. If we ignore it, we plan for a response that does not actually function when activated."

C.4 The Institutional Consolidation Finding ("Everyone Wears 15 Hats")

The institutional consolidation finding was articulated in conversation between the Launch Alaska Chief Deployment Officer and the author. The framing is attributed to role per the Chatham House convention.

Conversation between Launch Alaska Chief Deployment Officer and the author, during a wargame break.

Launch Alaska Chief Deployment Officer: "In Alaska, everyone wears 15 hats. The same small group of people staffs multiple critical functions. The Coast Guard sector commander was a marine inspector before that. The CISA advisor was a state emergency manager before that. The municipal emergency manager was a regional emergency manager in another jurisdiction before that. We all know each other. We have all worked together across multiple emergencies. This is what makes the response work under most conditions. It is also what makes the response fragile when those people are not available. You are not losing redundant personnel. You are losing people who each cover three or four critical functions."

[The framing was incorporated into the report's analytical treatment of how thirty-five to forty percent absenteeism produces larger operational consequences in Alaska than in equivalently-sized Lower 48 jurisdictions.]

C.5 Direct Labor as Essential Personnel

The direct labor finding extended from multiple participant exchanges across Move 1, Move 2, and Move 3. The cumulative point is that "first responder" as a Lower 48 construct does not capture the workforce categories Alaska's response architecture depends on emerged through repeated references to specific direct labor categories whose absence would stop specific operational functions rather than articulated as a single statement.

Day 2 morning, discussion of port operations under degraded staffing conditions.

Port of Alaska Operations and Security Support: "The gantry cranes need operators. The roll-on/roll-off ramps need operators. The fuel terminal valve operations need operators. None of this is automated. None of this can be done remotely. If we lose the longshoremen and stevedores who actually do the work, the port stops functioning. It does not matter what the response architecture looks like elsewhere. The port stops."

Day 2 afternoon, discussion of road maintenance during sustained ashfall and winter conditions.

Municipality of Anchorage Unified Command EOC Director: "Our snow removal capacity follows priorities. Main arterials first, then secondary roads, then neighborhoods. In a sustained event, residents in some areas are waiting through accumulating snowfall for days. There is no surge force for snow removal. The Lower 48 mutual aid that exists for fire response does not exist for snow removal because no other state has the equipment, the operator skill, or the cold-weather operational capability at the relevant scale."[91]

Day 1 afternoon, discussion of fuel distribution under degraded port conditions.

DLA Energy Quality Assurance Representative: "Our fuel distribution plan depends on a small number of trucking companies. Shoreline, Weaver Brothers, Carlile. These are specific companies with specific driver rosters. The drivers have commercial driver licenses with hazmat endorsements and Alaska-specific winter driving experience. You cannot substitute Lower 48 truckers for these drivers. The Alaska Highway is a different operating environment. The Tok Cutoff in February is a different operating environment. The drivers we need are the drivers we have."

[These exchanges and parallel ones across the wargame produced the finding that the responder category must extend to include the direct labor workforce that operates the physical infrastructure.]

C.6 The Lower 48 Surge Support Assumption

The Lower 48 surge support assumption operated implicitly through most of the gameplay. The exception that proves the rule emerged during Day 2 morning when participants briefly questioned whether eMAC requests for cold-weather National Guard personnel would actually be honored. The participants moved on from the question without resolving it. The post-game engagement identified that the assumption deserved direct treatment as a finding rather than as an unmodeled assumption.

Day 2 morning, brief discussion during Move 2 setup.

Alaska National Guard Representative: "We can put in eMAC requests to Minnesota, Michigan, Wisconsin. They are cold-weather states with the kit and the personnel. We would expect those requests to be honored."

USCG Fellow at Ted Stevens Center: "What happens if Minnesota has a concurrent crisis? What happens if we have multiple crises simultaneously across the country and the eMAC framework is contested?"

[91] Statement by participant from the Municipality of Anchorage Office of Emergency Management, Day 2 afternoon, April 2026. The snow removal limitation was discussed during the Move 3 wrap-up.

[Brief acknowledgment that the question was real, followed by movement to other topics. The post-game engagement identified this as the assumption the wargame did not adequately stress test.]

C.7 Canadian Unilateral Deployment and the NORAD Gap

The Canadian Navy HMCS Vancouver deployment was a white cell inject during Day 2 morning that showed the finding about NORAD's maritime mission gap. The participant response to the inject articulated the finding directly.

Day 2 morning, white cell inject of HMCS Vancouver deployment from Esquimalt.

Alaska National Guard representative: "HMCS Vancouver is coming through unilaterally. This is not coming through NORAD. NORAD does not have a standing maritime mission. The joint command is configured for counter-air-breathing threats. For a maritime crisis with strategic competition dimensions, even close allies have to act on their own authority because the alliance structure does not accommodate the requirement."

USCG Sector Western Alaska representative: "We welcome the Canadian deployment. We need the capability. But the institutional path matters. If we are building Arctic homeland defense architecture for the contemporary threat environment, NORAD as currently constituted is not the right framework. We need something that addresses maritime mission requirements explicitly."

C.8 The Chinese Coast Guard Marine Casualty

The Chinese Coast Guard vessel Nansha marine casualty in the ash plume was adjudicated through dice during Move 2 on Day 2 morning. The dice roll came up 8 of 12, indicating breakdown (1-8 producing the casualty). The subsequent discussion produced the report's finding about how environmental conditions degrade adversary capability and create U.S. operational opportunities.

Day 2 morning, white cell adjudication of the Nansha approach.

White Cell (Idaho National Laboratory Senior Critical Infrastructure Analyst): "Rolling for whether the Nansha suffers a marine casualty in the ash plume. The roll is 8. That is a casualty. The Nansha is dead in the water in the eastern Aleutians."

USCG Sector Western Alaska representative: "We send capability to render aid. This is what we do. The Chinese vessel needs rescue, we provide rescue. The strategic implication is that the Chinese humanitarian justification collapses. They came to provide humanitarian assistance and they needed rescue from us. That story writes itself in the information environment."

Alaska National Guard Representative: "They don't have operational performance in these conditions. We're going to have to go save their ass. This is the U.S. advantage. Not absolute platform survivability. Rescue and recovery capability in degraded operating environments."

[The exchange produced the finding about the strategic implications of U.S. rescue capability in adversary-degraded operating conditions.]

C.9 The Russian Icebreaker Approach and Tribal Aid Acceptance

The Russian icebreaker repositioning toward Nome and Kotzebue with humanitarian framing was the most significant Red Team-driven scenario element. The adjudication produced the report's finding on differential tribal community response to adversary humanitarian aid.

Day 2 morning, white cell adjudication of Russian icebreaker scenario.

White Cell: "Russian icebreakers are repositioning toward Nome and Kotzebue with humanitarian assistance framing. They are invoking Chukotka cultural ties, Russian Orthodox religious heritage, and historical relationships with Alaska Native communities. They are offering medical supplies, food, and shelter. Rolling for subsistence hunt success in the affected communities prior to icebreaker arrival. Roll is 9 out of 12. Hunts were adequate, communities are not facing immediate food crisis. Rolling for whether any community members board the Russian vessels. Roll is 3 out of 12. The threshold is 1-2 for yes; no community members board."[92]

USCG Sector Western Alaska representative: "What about aid offered without boarding? Can communities accept aid that the Russians off-load to the ice without anyone boarding the Russian vessels?"[93]

White Cell: "The legal authority to prevent aid acceptance beyond 12 nautical miles does not exist. Federal refusal does not bind community-level acceptance. If communities want to accept aid offered without boarding, we treat that as effectively certain."

USCG Sector Western Alaska representative: "If they want to walk out to the ice and accept the aid, we don't stop them. We facilitate the safety and security of the transfer. We make sure no Russians come ashore. But community-level acceptance of aid offered without boarding is not something we have the authority to prevent."

[The exchange produced the findings on differential tribal community response and on the limits of federal authority over community-level aid acceptance.]

C.10 The Red Team's Strategic Logic on Cost-Imposition and Information Operations

The Red Team's most significant contributions came during the Day 2 afternoon hot wash, when the Red Team articulated its strategic logic for the moves it had chosen and the moves it had deliberately chosen not to execute. The Red Team's framing supports findings on cost-imposition asymmetry.

Day 2 afternoon, Red Team explanation during hot wash. Attributed to "the Red Team" collectively per the Red Team non-attribution agreement.

[92] Day 2 afternoon transcript, Red Team strategic framing during hot wash. The targeting of vulnerable populations with medicine for sick families was identified explicitly.
[93] Day 2 morning transcript, Russian icebreaker scenario discussion. Statement by participant from the U.S. Coast Guard.

Red Team: "We deliberately did not use cyber. The environmental degradation had already produced the operational effects that cyber action would have produced. Why reveal capability for redundant effect? We get cost-imposition for free from the volcano. We just need to add ambiguity at the margins."

Red Team: "The backpack incident at the Port of Alaska is the textbook case. We did not need to actually damage the valve. We just needed the defenders to lock in a disproportionate response the moment the ambiguous threat emerged. FBI engagement, MARSEC elevation, FPCON adjustment, personnel diversion. The cost is locked in at the moment of initial response. Whether the backpack actually contained explosives is irrelevant to the cost-imposition we obtained."[94]

Red Team: "The Russian Orthodox angle in Alaska Native communities is real. You only need a couple. You have a couple of villages, and especially in the more northern communities that are currently going to the UN panel because the federal government has not taken care of the contaminants. You have disenfranchised citizens that may create an opportunity. We are not making this up. The historical pattern is documented. The targeting is the standard playbook applied to a new geography."

Red Team: "What we are showing you is that the defender's institutional architecture is the variable we are actually targeting. Not the specific operational asset. Not the specific community. The architecture itself. How the FBI is prioritized. How MARSEC works. How the clearance system creates information asymmetry between cleared officials and the public. How the response architecture defaults to sequential rather than parallel processing. All of these are exploitable. We just need to generate ambiguity. The architecture does the rest."[95]

C.11 The Backpack Incident and FBI Prioritization

The backpack incident at the Port of Alaska's valve operating system, during Move 2 on Day 2 morning, produced the report's finding on FBI procedural primacy and the cost-imposition mechanism.

Day 2 morning, backpack incident sequence.

White Cell: "Backpacks have been spotted near the valve operating system at the second fuel terminal. Status unknown."

Anchorage Port of Alaska representative: "We elevate immediately. FBI engagement is the protocol. We are not making determinations about contents until the FBI arrives and clears the scene. The port operations in that area shut down pending FBI clearance."

[94] Day 2 morning transcript, backpack incident sequence. The full sequence of initial alert, FBI engagement, MARSEC elevation, FPCON adjustment, eventual determination of non-explosive nature is captured in the transcript. The cost-imposition mechanism the incident demonstrates is the textbook case the report develops in subsequent analysis

[95] The "clearance as chaos" framing is the lead authors' shorthand for the information asymmetry the wargame illuminated.

Port of Alaska Operations and Security Support: "That terminal is one of two operational at this point. The other one was damaged in Move 1. If we shut down operations at this terminal for FBI clearance, we are at single-terminal capacity for the duration."

USCG Sector Western Alaska representative: "MARSEC level elevation is automatic given the credible threat indicator. FPCON adjustment follows. We are now in a posture that requires sustained security augmentation that we do not have personnel for given the absenteeism conditions."

[White Cell adjudication subsequently determined that the backpacks did not contain explosives, but the security posture elevations that the initial response triggered did not unwind. The exchange produced the report's finding that ambiguous low-cost actions force high-cost institutional responses regardless of whether the action turns out to be a real threat.]

C.12 The "Clearance as Chaos" Information Asymmetry

The clearance-uncleared information asymmetry was visible across multiple sessions but emerged most explicitly during Day 2 morning when cleared participants were aware of intelligence indicators that uncleared participants in the same room did not have access to.

Day 2 morning, during discussion of subsea cable disruption attribution.

USCG Sector Western Alaska representative: "I can speak to what I can speak to in this forum. The classified picture includes elements that are relevant to the question of attribution but that I cannot share with everyone in the room."

DHS CISA Protective Security Advisor: "This is the operational reality. Cleared officials know things that uncleared officials in the same response operation do not know. The public knows even less. The standard public affairs frameworks assume this asymmetry does not produce operationally significant problems. The wargame is demonstrating that it does."

Municipality of Anchorage Unified Command EOC Director: "When we are running the response, we are operating on what we know. If the cleared officials know more about the threat picture than we do, our response decisions are inevitably less informed than they could be. This is not a complaint. It is a condition of how the clearance system intersects with operational response."

C.13 Sequential versus Parallel Planning

The sequential-versus-parallel finding emerged through gameplay observation across multiple sessions and was articulated most clearly in one of the military participants' after-action submissions. The supporting transcript material reflects the gameplay pattern that produced the observation.

Day 1 afternoon and Day 2 morning, across multiple discussions of decision sequencing.

[Pattern observed across multiple sessions: participants worked through DSCA activation, then comms issues, then aviation assumptions, then Alaska Marine Highway retasking as a sequence. Each was identified as a concern relatively early in the gameplay. Each was executed substantially later than

identified. The pattern of identifying issues early and executing responses late was visible to participants in real time but did not produce real-time correction. The post-gameplay reflection identified this as the central institutional tempo finding.]

One of the military participants, in his after-action submission: "The most consequential learning from our decision-making process was the cost of sequential rather than parallel planning. The DSC activation, the comms issues, aviation optimism assumptions, and the Alaska Marine Highway retasking were all identified early and executed late. Each delay in a real situation would have compounded. The wargame demonstrated clearly that information operations, command architecture, and logistics must be stood up simultaneously on Day 1 as there is no sequential phase in which it is safe to defer any of them."

One of the military participants, in her after-action submission: "Information sharing by white cell design and within our self-imposed silos was the challenge today, especially as geography and concurrent events were occurring. Group dynamics were significantly more productive today, but we did have twoish clear silos of effort and we struggled to bridge the gaps. I am guessing that is a very likely scenario."

C.14 Weather Data as Operational Input

The weather data critique emerged from one of the military participants' after-action submission and supports the methodology finding on the design oversight in providing only summary weather information rather than operationally-specific data.

One of the military participants, in her after-action submission: "As an operator, I really struggled with the lack of weather information (both the immediate and general forecasts) provided by white cell, especially as the driver of the scenario was a natural disaster where the consequences of the eruption would have been affected by Meteorological phenomena. I didn't argue it during the scenario, but I think response challenges here in AK are as much responding to the casualty as managing the weather (and the weather forecast). Providing a couple local area forecasts or NOAA weather charts would have grounded the scenario into reality a little more. Temperature, wind, sea state, and visibility at sea level and the different aviation levels would have been great!"

C.15 The Geographic Heterogeneity of Rural Alaska

The geographic heterogeneity finding emerged from one of the military participants' after-action submission and supports findings on the differential needs across rural Alaska communities.

One of the military participants, in her after-action submission: "I think we severely underestimated how geographically varied the different surrounding community needs would be and how those would have impacted our overall operations. Not sure if the local EOC discussed community services like medivacs [sic] or state trooper support for crime but Anchorage is the health care hub, and our rural villages are often without their own police force and require troopers to travel in/out. Those demands would have been competing with other demands and required some reallocation of assets."

C.16 Summer Variant as Follow-on Recommendation

One of the military participants, in her after-action submission: "This was a fun scenario in the winter, but I think it would be a really interesting summer time problem. The sheer amount of extra people in Alaska (ex. tourists, outdoor recreationists, commercial fishermen, seasonal and construction workers) would change the resourcing for response efforts, not to mention the increased maritime hazards with more vessels operating and then the follow on economic consequences from impacts to all the seasonal industries. Just my own thoughts but I think the cold weather here often makes things easier."

C.17 Volunteerism Research as Follow-on Question

One of the military participants, in her after-action submission: "I would love to be a fly on the wall when State and various local officials (to include more rural communities) are working alongside Federal Agencies and how adding State changes the dynamic. I would be curious if we as a group overestimated rural resilience or if we underestimated it. I would also be curious how the volunteerism levels are in actuality, in different locations and how those organic efforts would inform emergency management officials both for the natural disaster and as it pertains to intelligence gathering and to the willingness of the EOC to accept volunteers or intel."

C.18 The Traffic Cones Approach to Evacuation

The Anchorage emergency management approach to evacuation through traffic management rather than law enforcement deployment was articulated by the Municipality of Anchorage Unified Command EOC Director during Day 1 afternoon. The approach reflects her professional operational practice. The wargame co-lead's reservation about the capacity of Alaska road infrastructure is incorporated into analysis as productive engagement between operational best practice and reality.

Day 1 afternoon, discussion of evacuation planning approach.

Municipality of Anchorage Unified Command EOC Director: "We would not be using police officers for traffic control in an evacuation situation. When people see law enforcement officers, it slows everything down. People stop and ask questions. People worry they are in trouble. Traffic cones and barriers move traffic. Police officers stop traffic. Our approach is traffic management through physical infrastructure rather than through personnel."

APPENDIX D | EXECUTION LESSONS

Several execution choices warrant honest examination.

The decision to provide minimal upfront ground-truthing at the start of Move 1 was deliberate. The designers wanted to test how participants would self-organize when forced to establish the operating picture themselves. The cost was that Move 1 spent significant time on baselining rather than on responding to injects. The benefit was that the gameplay showed exactly which assumptions participants would and would not arrive at independently. Both Alaska National Guard and USCG representatives identified this trade-off in their after-action submissions, with both recommending that future iterations be built in a more upfront context. The report accepts this recommendation while noting that the open-ended approach produced specific findings, including the natural-gas-turbine vulnerability, that more scripted approaches likely would not have exposed.

The decision not to provide weather data at the level of fidelity Alaska operators actually work with was a design oversight rather than a deliberate choice. USCG representative's after-action submission identifies this directly: "As an operator, I really struggled with the lack of weather information (both the immediate and general forecasts) provided by white-cell, especially as the driver of the scenario was a natural disaster where the consequences of the eruption would have been affected by meteorological phenomena. Response challenges here in AK are as much responding to the casualty as managing the weather and the weather forecast." The recommendation she articulates, "that future iterations provide local NOAA charts, temperature, wind, sea state, and visibility at sea level and different aviation levels," is one this analysis endorses without qualification. Weather is a primary operational input in Alaska, not scenario backdrop, and the wargame design should have treated it as such.

The decision to introduce the gray-zone dimension only in Move 2 was a design choice the report now identifies as one that should be reconsidered for future iterations. The military fellow representative's after-action submission articulates the critique: "The grey zone dimension also arrived late in game flow; introducing it as a background condition from Move 1 would better reflect the real operational environment and give the HADR-focused audience more time to grapple with how security concerns constrain humanitarian choices." The wargame's central finding, that strategic competition is the persistent operating environment within which Arctic HADR now occurs, is in tension with a design that delays the gray-zone dimension to a later move. Future iterations should introduce gray-zone activity as a Move 1 background condition.

The use of dice to adjudicate uncertain outcomes worked when it was used. Several participants in the hot wash recommended more extensive use of dice rather than discussion to resolve specific questions, on the grounds that real institutional response often makes decisions under uncertainty without time for deliberative consensus.

The wargame's dice rolls produced specific findings: the subsistence-hunt-pass roll; the no-personnel-board-the-Russian-icebreakers roll, the community-aid-acceptance-without-boarding effective certainty, the Chinese-cutter-marine-casualty roll, that shaped subsequent gameplay. Future iterations should default to dice adjudication more readily.

Boundary conditions on findings.

- The wargame produced multiple categories of analysis, and the report distinguishes among them explicitly.
- Observations are claims the wargame directly produced such as institutional behaviors, decisions made under stress, adjudicated outcomes through gameplay.
- Hypotheses are claims the wargame raised but did not establish. The differential tribal response finding is a hypothesis. The "no-scalable-alternatives" framing as applied to specific infrastructure investments is a hypothesis informed by the wargame and by external evidence. These are reported as hypotheses warranting follow-on investigation.
- Recommendations are claims the convergence of evidence supports, the wargame plus external research, expert judgment. The two-deep-water-ports-and-two-airports recommendation is a recommendation rather than a wargame finding in the narrow sense. The wargame supports the underlying claim about hub vulnerability; the specific port and airport recommendations rest on

the convergence of the wargame finding with existing Army Corps of Engineers feasibility work on Nome and Port Clarence and with Congressional Research Service analysis of Arctic infrastructure requirements.

- Questions are gaps the wargame opened but could not resolve. The actual volunteerism dynamics in different Alaska communities. The summer-scenario vulnerabilities that the winter design did not test. The integration of Alaska Native corporations into emergency response architecture. The functional realities of the Joint Information Center concept under conditions of degraded communications. These are reported as questions warranting follow-on work.
- The wargame is not a controlled experiment and does not produce findings of that character. It is a structured exploration of how an interconnected set of institutional actors responds to a compound stressor scenario, with adjudication mechanisms designed to engage dynamics that purely deliberative exercises would suppress. Its value lies in what it raises and more research is needed to further prove findings. The report is careful throughout to distinguish among these categories so that readers can assess the conclusions on appropriate evidentiary grounds.

APPENDIX E | Glossary of Acronyms and Alaska-Specific Terms

This glossary defines the acronyms and Alaska-specific terms used. Federal and military acronyms are listed first, followed by Alaska-specific organizational acronyms and Alaska-specific operational terms.

Federal and Military Acronyms

AAR — After-Action Report. A document produced after an exercise or operation that captures lessons learned, observations, and recommendations.

ALCOM — Alaskan Command. The U.S. Indo-Pacific Command sub-unified command headquartered at Joint Base Elmendorf-Richardson with operational responsibility for U.S. military forces in Alaska.

ATC — Air Traffic Control.

CBRN — Chemical, Biological, Radiological, and Nuclear (threats and response capabilities).

CFR — Code of Federal Regulations. The codification of federal administrative law.

CINC — Composite Indicator of National Capability. A measure of national power used in international relations research, aggregating military expenditure, military personnel, energy consumption, iron and steel production, total population, and urban population.

CISA — Cybersecurity and Infrastructure Security Agency. A component of the Department of Homeland Security with responsibility for critical infrastructure protection.

CST — Civil Support Team. A National Guard unit with specialized hazmat and chemical, biological, radiological, and nuclear response capabilities, operating under Title 32 authority.

DHS — Department of Homeland Security.

DLA — Defense Logistics Agency. The Department of War combat support agency responsible for supply chain management.

DoW — Department of War.

DOJ — Department of Justice.

DOT — Department of Transportation.

DSCA — Defense Support of Civilian Authorities. The framework for U.S. military support to civilian authorities during domestic emergencies.

DTRA — Defense Threat Reduction Agency.

eMAC — Emergency Management Assistance Compact. The interstate mutual aid agreement among states for emergency response support.

EOC — Emergency Operations Center.

EWR — Early Warning Radar.

FAA — Federal Aviation Administration.

FBI — Federal Bureau of Investigation.

FEMA — Federal Emergency Management Agency.

FMSC — Federal Maritime Security Coordinator. The U.S. Coast Guard official with operational authority over maritime security in a defined sector.

FPCON — Force Protection Condition. A Department of War scale of threat levels, from Normal through Alpha, Bravo, Charlie, and Delta, that determines security posture at military installations.

GAO — Government Accountability Office.

GBMD — Ground-Based Midcourse Defense. The U.S. ballistic missile defense system that intercepts intercontinental ballistic missiles in the midcourse phase of flight. Approximately ninety percent of the deployed GMD interceptors are at Fort Greely, Alaska.

HA/DR — Humanitarian Assistance and Disaster Response.

ICE — U.S. Immigration and Customs Enforcement.

ICS — Incident Command System. The standardized hierarchical structure used for emergency response in the United States.

INL — Idaho National Laboratory. A Department of Energy national laboratory with infrastructure and security analysis capabilities.

IPCC — Intergovernmental Panel on Climate Change.

JBER — Joint Base Elmendorf-Richardson. The principal U.S. military installation in Alaska, hosting Air Force, Army, and headquarters elements of Alaskan Command.

JIC — Joint Information Center. A coordinated public affairs center used during major emergencies to manage information flow to the public.

LNO — Liaison Officer.

MARSEC — Maritime Security Level. A U.S. Coast Guard scale of threat levels from MARSEC 1 (lowest) through MARSEC 3 (highest) that determines security posture at port facilities and on vessels.

MOTR — Maritime Operational Threat Response. An interagency coordination mechanism for maritime threats.

NDS — National Defense Strategy.

NEMA — National Emergency Management Association.

NIPP — National Infrastructure Protection Plan. The Department of Homeland Security framework for critical infrastructure protection.

NORAD — North American Aerospace Defense Command. The bi-national U.S.-Canada command responsible for aerospace warning, aerospace control, and maritime warning for North America.

NORTHCOM — U.S. Northern Command. The U.S. unified combatant command with responsibility for homeland defense in the continental United States, Alaska, Canada, and Mexico.

NRI — National Risk Index. The Federal Emergency Management Agency's community-level risk assessment tool.

NSAR — National Strategy for the Arctic Region. The Biden administration's 2022 Arctic strategy.

NSS — National Security Strategy. The administration-level strategic guidance document, most recently issued in 2026.

NSR — Northern Sea Route. The maritime route along the Russian Arctic coast.

PSA — Protective Security Advisor. A Department of Homeland Security CISA field representative.

RAND — RAND Corporation, a federally funded research and development center.

SSI — Sensitive Security Information. A category of unclassified but protected information.

TFR — Temporary Flight Restriction. A Federal Aviation Administration mechanism for restricting airspace use during specific events or operations.

TSC — Ted Stevens Center for Arctic Security Studies. The Department of War's regional center for Arctic security education and research, located at Joint Base Elmendorf-Richardson.

UAS — Unmanned Aircraft System. Also referred to as drones.

USACE — U.S. Army Corps of Engineers.

USCG — U.S. Coast Guard.

VOAD — Voluntary Organizations Active in Disaster. The national association of major nongovernmental disaster response organizations.

Alaska-Specific Organizational Acronyms

ACEP — Alaska Center for Energy and Power, at the University of Alaska Fairbanks.

ACUASI — Alaska Center for Unmanned Aircraft Systems Integration, at the University of Alaska Fairbanks.

AFN — Alaska Federation of Natives.

AIDEA — Alaska Industrial Development and Export Authority.

AK NG — Alaska National Guard.

ANCSA — Alaska Native Claims Settlement Act of 1971.

APD — Anchorage Police Department.

AVO — Alaska Volcano Observatory.

MOA — Municipality of Anchorage.

TAPS — Trans-Alaska Pipeline System. The 800-mile pipeline carrying crude oil from Prudhoe Bay on the North Slope to the marine terminal at Valdez.

UAF — University of Alaska Fairbanks.

Alaska-Specific Operational Terms

The Alaska Highway — the principal ground transportation route between Alaska and the continental United States, running through Canada.

Bering Strait — The strait between Alaska and the Russian Far East (Chukotka), approximately 55 miles wide at its narrowest point.

Bypass mail — The U.S. Postal Service program that delivers goods to rural Alaska communities not accessible by road, treating mail and freight as integrated cargo categories.

Bristol Bay — The southeastern arm of the Bering Sea, the location of the world's largest sockeye salmon fishery and the focus of substantial seasonal Alaska economic activity.

Chukotka — The Chukchi Autonomous Okrug of the Russian Far East, directly across the Bering Strait from Alaska, with historical and cultural ties to Alaska Native communities.

Cook Inlet — The body of water extending south-southwest from Anchorage, providing access to the Gulf of Alaska. The Cook Inlet natural gas production infrastructure supplies the Anchorage region's electrical generation.

Glenn Highway — The principal road connecting Anchorage to the interior of Alaska, including Glennallen and the route to the Alcan.

Marine Highway System — The Alaska state ferry system, operating between communities along Alaska's coastline and providing maritime transportation for communities not accessible by road.

Mount Spurr — The volcano approximately 80 miles west of Anchorage that has historically produced ash events affecting the Anchorage region. Referenced in the wargame for its operational precedent for family evacuation patterns.

Pribilof Islands — A group of islands in the Bering Sea, the historical center of the Aleut/Unangan population and a focus of Russian Orthodox religious heritage in Alaska.

Seward Highway —The only highway out of Anchorage, it extends north to the Glenn Highway and south to the Kenai Peninsula.

Tok Cutoff — The road junction in interior Alaska connecting the Glenn Highway to the Alaska Highway, a critical chokepoint for ground evacuation from south-central Alaska.

APPENDIX F | After Actions Summaries | Consolidated Feedback Forms

Separate Document as participate's names are on forms.

APPENDIX G | Consolidated Bibliography

This bibliography consolidates all sources cited in the report. The dissertation cited as the primary analytical source is listed first; all other sources follow in alphabetical order by author or institutional issuer within thematic groupings.

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